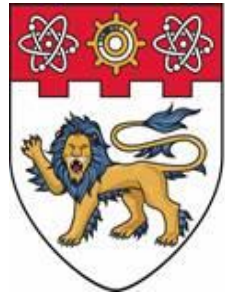


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OPTIMAL FOIL GEOMETRY FOR FLOW ENERGY HARVESTING

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OPTIMAL FOIL GEOMETRY FOR FLOW ENERGY HARVESTING

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BY

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ABSTRACT

In recent years, there has been an increasing demand for renewable and sustainable energy technological advancements. This has inspired an interest in bio-inspired energy harvesting systems, among which the flapping foil mechanism offers a promising potential. It offers the means of extracting energy from fluid flow, however, the influence of the geometry of the foil, particularly non-traditional shapes and variations in thickness, remains largely unstudied.

This study investigates how the geometric shaping and thickness of foils in a flapping foil system affects the wake dynamics and consequently, the performance of the system in harvesting energy under low-speed, in-line tandem configuration. A total of 9 pairs of Polylactic Acid (PLA) foils were 3D printed, with 1 pair being a baseline, geometrically flat and thin foil. The foils are separated into 2 categories: (1) **Winglet**, where the edges are shaped with a largely flat mid-section, (2) **Geometric**, where the foils follow a consistent shaping throughout. The foils were tested in a recirculating water tunnel operating at a flow velocity of 0.15 m/s . Laser sensors were mounted to obtain displacement-time data which is later processed through FFT methods to obtain data such as frequency, Strouhal number (St), power (P) and efficiency (η). Hydrodynamic forces such as lift and its corresponding coefficient of lift (C_L) were estimated through indirect damping-based models.

The results achieved in this study demonstrated that both the geometric design and thickness of the foil have strong influences over the performance. The foils in the geometric category were observed to consistently outshine the foils in the winglet category, achieving higher numbers across all performance metrics – higher Strouhal numbers of up to $St = 2.5$ and efficiencies of up to 45%. It was also found that optimal performances occurred within the 5 – 10mm window, indicating that moderate thickness is a key for enhanced energy harvesting. On the other hand, it was noted that the rectangular foils demonstrated consistently below average performances, often showing declining trends that peak below the baseline values across all metrics. The force or lift based metrics also exhibited significant noise due to indirect methods of reconstruction and estimation alongside the setup having increased sensitivity to noise due to low freestream velocity.

The findings in this study highlight the effect of fully shaped geometries of foils in a flapping-foil system and how it tends to enhance hydrodynamic performances in unsteady flow. Having a smoothly curved or tapered geometry (e.g. circular, elliptical, triangular) tend to promote more stable vortex formations and increased speeds of oscillation, while the opposite occurs with blocky, highly angular geometries (e.g. rectangular). The results also align with existing studies on bio-inspired propulsion or flapping foil mechanisms and provide useful insights on the potential designs for future applications in energy harvesting devices.

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1. INTRODUCTION

1.1 Background

In recent years, climate change and global warming have been a concern and a focus of study. As such, the demand for sustainable and eco-friendly energy sources has increased globally. To preserve the health of the environment, a significant number of studies and research have been conducted to tackle the issue, shifting from traditional sources of energy (e.g. fossil fuel) to developing renewable energy sources [1]. The primary sources of sustainable energy can be found in solar, wind, tidal, and more, each offering their respective benefits and disadvantages in implementation. Among these, both wind and tidal sources show promising results and potential when it comes to harvesting renewable energy. According to a report by the International Energy Agency (IEA), a rapid scale-up of wind energy capacity of 11-fold by 2050 is being pushed for [2]. A separate report by Ocean Energy System (OES) projects to develop 300GW of ocean energy by 2050 [3].

Lately, researchers have taken a keen interest in observing aerial and aquatic wildlife, drawing inspiration from their excellent propulsion performances. This led to the rise of interest in bio-inspired designs being developed used to harvest energy such as hydrofoils that can turn unsteady flow into energy through oscillations [4]. As mentioned, the development of these bio-inspired propulsion systems looks to nature for inspiration and insight into maximizing the efficiency of motion through fluids. Flapping foils – structures that utilise heave and pitch oscillations to produce thrust – are promising developments in energy harvesting as well as marine propulsion [5], [6]. The mechanism behind flapping foils imitates that of aquatic and aerial life, such as fins and insect wings, respectively. These optimise unsteady flow to generate lift and thrust more effectively than the current methods of rotary propulsion systems at low Reynold numbers [7], [8].

Branching from this, the tandem configuration which involves 2 flapping foils in a series alignment placed along the direction of flow, is explored. The foil in the downstream functions off the wake of that in the upstream. Depending on the alignment, spatial and temporal, the interaction between both foils and the vortices released can either impede or augment the performance [9], [10]. Configurations like this can potentially allow for better performance through optimal wake interferences. This, however, requires 2 factors namely phase lag and foil spacing to be optimised [11], [12].

Current research into tandem foil usually involves symmetric airfoils that have sinusoidal kinematics. While these have been well-researched, natural motion may appear more complex and therefore have been unable to capture the full picture [13], [14]. Essentially, wake produced by rigid, symmetric foils differ from that of bio-inspired foils that tend to lean more toward flexibility and asymmetry (e.g. palm grass leaf).

This study intends to explore wake characteristics exhibited by palm grass foils. Palm grass is observed to display large amplitude motions, varying thickness and delayed stroke reversals. Among these, a variable that is often overlooked, the varying thickness

could affect vortex shedding, flow separation and overall coherence of wake structures [15] [16].

1.2 Objective

Even though tandem wake interactions have been modelled using linear potential theory or vortex methods, there are few experimental studies that examine the effects of geometric variations, such as thickness and shape, on the configuration's performance. This presents a gap in current literature and motivates the need for more data-driven studies that incorporate non-traditional foil designs under controlled laboratory conditions.

This project aims to investigate the foil-wake interaction of palm grass foils of varying thickness and airfoil shapes in a tandem configuration. The objective is to characterize the resulting wake fields and their influence on downstream foil performance, with implications for future designs in energy-efficient propulsion and bio-inspired systems.

1.3 Scope

The scope of the project is outlined below to achieve the abovementioned objectives:

- Design with Computer Aided Design (CAD) airfoils of varying geometry – shape and thickness -- then printing of airfoils.
- Utilising a water tunnel, measure the flapping velocity and acceleration of the different models.
- Collect the data using laser sensor that is mounted to measure the distance travelled by the test model.
- Using MATLAB, process the collected data to calculate the energy efficiency of each airfoil geometry.
- Comparing the results, determine the optimal geometry for energy harvesting.

1.4 Outline

In Chapter 2, literature reviews of past studies and experiments will be done. The characteristics of tandem configuration, flapping foils and palm grass foils will be examined and discussed. In Chapter 3, the methodology of this study will be described, highlighting the tools and methods used to conduct and evaluate this project. In Chapter 4, the data collected will be evaluated and compared along with derivations and equations involved. The comparisons will shed light on the optimal geometry of airfoil with respect to energy harvesting. Lastly, in Chapter 5, conclusions will be made for this study, discussing current limitations and possible future work to be followed up upon.

2. LITERATURE REVIEW

2.1 Flapping Foils

2.1.1 Kinematics of Flapping Foils

The kinematics of flapping foils involve 2 oscillatory motions, namely heave and pitch, to produce thrust. Heave refers to vertical movement while pitch refers to the rotation about an axis or point. The parameters and non-dimensional groups used are:

- Heave / Vertical Displacement:

$$y(t) = y_0 \sin(2\pi ft) \quad (1)$$

- Pitch angle:

$$\theta(t) = \theta_0(2\pi ft + \phi) \quad (2)$$

Where y_0 is the heave amplitude, θ_0 is the pitch amplitude, f is the frequency and ϕ is the phase lag [17], [18].

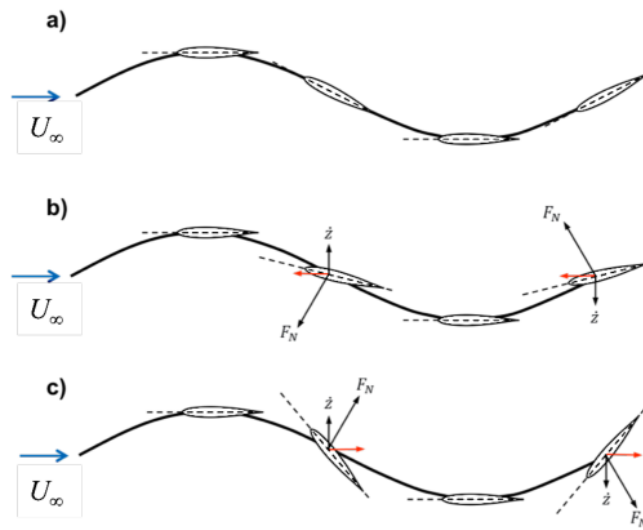


Figure 1 Schematic of flapping foil heave and pitch [19]

Referring to Figure 1, the heaving and pitching of the foil generates a time-varying normal force (F_N). The streamwise component of F_N contributes to thrust production. Proper tuning of the amplitude, frequency, and phase lag enhances the formation of a reverse von Kármán wake, which is associated with net thrust generation. The timing between heave and pitch affects how effective the alignment of the foil is with the induced flow, thus influencing the strength and direction of the thrust vector.

Next, the Strouhal number (St) is a dimensionless parameter used to analyse the propulsive efficiency of flapping foils, more specifically towards bio-inspired flapping foils.

$$St = \frac{2y_0 f}{U} \quad (3)$$

Where U is the freestream velocity. Typically, propulsion is at its optimal when the value of St is within a certain range:

$$0.25 \leq St \leq 0.35 \quad (4)$$

The above is largely supported by biological observations and experiments [11], [4], [17].

Third, is reduced frequency. For this study, it refers to the parameter that illustrates the ratio of inertial to elastic forces, or to be precise, the ratio of oscillation frequency to the velocity of flow. It is defined as such:

$$k = \frac{\pi f c}{U} \quad (5)$$

Where c is the chord length. It is observed that as the value of reduced frequency increases, it results in the loss of power output that comes from the angular momentum [20]. The reduced frequency is also related to the Strouhal number, where both are used to analyse kinematics of unsteady flow and wake structures [17].

$$St = \frac{kA}{\pi c} \quad (6)$$

Lastly, phase lag (ϕ), which refers to the offset time between the heaving and pitching motion and affects the overall propulsive performance. When there is optimal phase lag between the heaving and pitching, it allows the foil to align effectively with the relative flow and therefore, improves the thrust produced and efficiency.

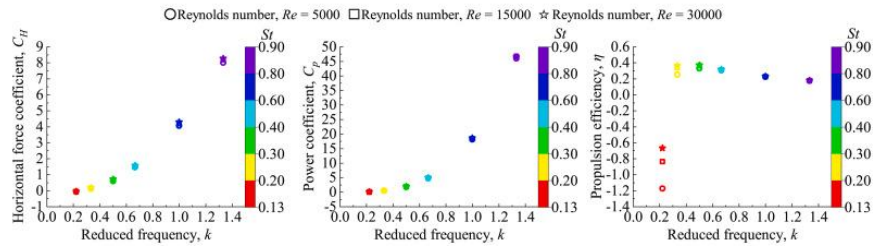


Figure 2 Horizontal force coefficient, power consumption coefficients and propulsion efficiency for various Strouhal numbers by changing the k value and maintaining a constant h value [21]

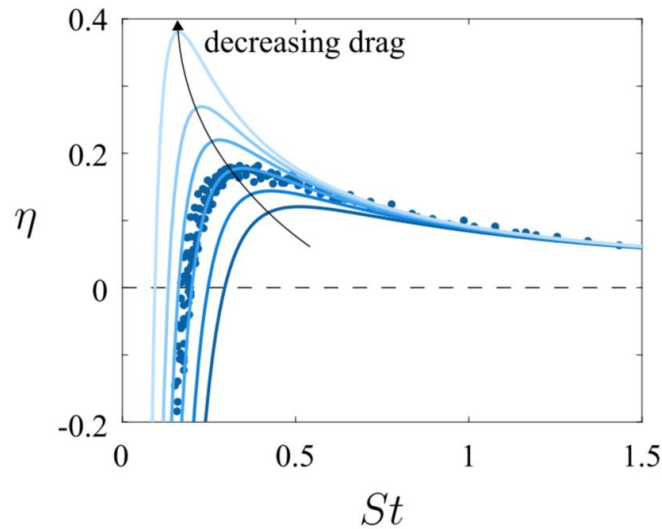


Figure 3 Relationship between Propulsive Efficiency (η) and Strouhal Number (St) [22]

In general, with an increased phase lag, there is an increase in performance in propulsion. However, past a certain optimal range or value, the performance will decline or plateau. With respect to tandem configurations, phase lag also affects how the trailing foil relates and interacts with the upstream wake. Research has shown that having an appropriate phase lag leads to a significant improvement in thrust produced as well as reducing the loss of energy [17], [18].

2.1.2 Applications in Nature vs Engineering

In nature, many parallels can be found for flapping foils, and many have led to many engineering applications. The oscillating foil is employed in biological systems in the form of both aquatic and aerial animals such as birds, fishes and insects. This is done to be able to achieve higher locomotion efficiency in unsteady environments (i.e. unsteady flow) [4], [11], [18]. Observed in these animals are coordinated heaving and pitching being utilised to generate lift and / or thrust. Additionally, they operate at an optimal Strouhal number and range of reduced frequency for maximum efficiency. For example, fish are observed to exhibit tail-beat frequencies and amplitudes corresponding to the Strouhal number range of $0.25 \leq St \leq 0.35$. This results in the production of the characteristic reverse von Kármán wake associated with efficient propulsion [11], [23].

In engineering, flapping foils are used across multiple applications such as:

- **Marine:**

In marine propulsion, underwater vehicles and robots have been developed toward mimicking biological animals to replace or compliment the current conventional screw propellers. This offers better manoeuvrability and reduced noise [23], [24].

- **Renewable energy:**

New and improved foils are being developed to further enhance renewable energy harvesting. Oscillating hydrofoils are a key design to extract the energy

from tidal currents. They operate similar to how wind turbines harvest energy but instead, convert oscillatory motion into the necessary mechanical or electrical power [24], [25].

- **Aerospace:**

In aerospace and Unmanned Aerial Vehicles (UAVs), aerial animals are a source of inspiration for design. Micro Air Vehicles (MAVs) are deploying bio-inspired flapping wings to achieve aerial manoeuvres like hovering or flying in confined spaces. These mimic the flight of insects and small birds [4], [18].

The tandem foil configuration takes inspiration from that of nature, optimising the use of upstream wake. In nature, this principle is exploited by schools of fish or even migrating birds to conserve energy over long distances of travel. In the context of engineering, the tandem configuration allows for improved efficiency in multi-foil propulsion systems and could increase the power extracted in energy harvesting [25].

2.1.3 Metrics of Performance

Flapping foils are often measured by non-dimensional metrics, allowing easier comparison across varying shape, size, flow conditions and more. These metrics are derived from the forces and power generated by the foil.

- **Coefficient of Thrust (C_T)**

Defined as the thrust force divided (nominator) by dynamic pressure (denominator).

$$C_T = \frac{F_T}{0.5\rho U_\infty^2 cs} \quad (7)$$

Where F_T is the mean thrust, ρ is the fluid density, U_∞^2 is the freestream velocity, c is the chord length and s is the foil span [6], [16].

- **Coefficient of Power (C_P)**

Defined as

$$C_P = \frac{P}{0.5\rho U_\infty^3 A} \quad (8)$$

Where P is average power require to heave and pitch the foil, $A = c \cdot s$ is the area. This measures the non-dimensional input power [26], [5], [16].

- **Propulsive Efficiency (η)**

Defined as

$$\eta = \frac{F_T U_\infty}{P} \quad (9)$$

It describes the ratio of useful power – thrust multiplied by freestream velocity – over the input power. At optimal reduced frequency and phase angle, the efficiency can be maximised [6], [16].

- Strouhal Number (St)

As earlier defined in equation (3), the Strouhal number is a key factor and dimension when it comes to investigating bio-inspired foils and propulsion. Within the abovementioned range of $0.25 \leq St \leq 0.35$, thrust production and power used is optimally balanced [4], [11], [17], [23].

- Reduced Frequency (k)

This dimension as defined in equation (5) connects the frequency to the chord and freestream velocity. Generally, the higher the k value, the higher the lift generated in unsteady flow, with some exceptions of plateauing past certain values of reduced frequency. However, this general relationship can also lead to increased power requirements [23], [24].

- Effective Angle of Attack (α)

The effective angle of attack, α , refers to the combined factor of the pitch angle and the velocity of induced flow that comes from the heaving motion of the foil. Larger angles of attack can result in an increase of instantaneous thrust due to stronger vortex formations. Yet, too large of an angle of attack may cause dynamic stalls and therefore, a reduced efficiency [24].

Though these are commonly used to measure flapping foils in oscillatory motions, this study will focus on energy or propulsive efficiency (η) and the Coefficient of Power (C_P).

2.2 Tandem Configurations

2.2.1 In-line vs Staggered

Traditionally, tandem foil configurations can be classified broadly into two categories of configuration: in-line and staggered. The in-line configuration refers to when the trailing foil is directly downstream of or in-line with the leading foil in the flow axis. On the other hand, the staggered arrangement refers to when the trailing foil is laterally or vertically offset from the leading foil. This arrangement is meant to mimic the offset observed in schools of fish or flocks of birds [8], [23].

When it comes to in-line configurations, studies have shown that the spacing ratio, defined as span divided by chord (equation 10), and the phase lag strongly influences the performance of the foils. It is reported that at an operating Strouhal number of 0.3, up to a 50% increase in thrust is achievable just by having an optimally placed trailing foil as compared to the baseline single foil [7]. Building on that, another study found that the wake alignment of in-line foils could enhance or decrease performance, which highlights how sensitive the spacing between foils is [9]. Additionally, with proper phase coordination, it is numerically proven that these in-line configurations can exploit the upstream foil's reverse von Kármán wake which

results in an improved propulsive efficiency [10]. In a more recent study, experiments were conducted to confirm the results, showing that there were improvements in thrust generated of approximately 25% when the trailing foil operates with a phase lag of an estimated quarter cycle relative to the upstream foil [11].

$$\text{Spacing ratio} = \frac{S}{c} \quad (10)$$

Conversely, staggered configurations aim to mimic and replicate energetically favourable formations observed in nature. It is investigated that by vertically offsetting the foils, vortex interference can be reduced. As such, staggered formations are shown to benefit from partial wake shielding observed in the downstream foil, which can only be achieved in this configuration [13]. This means that thrust is maintained whilst lower drag is experienced and can therefore improve the performance stability of this configuration compared to in-line. These findings agree with nature's observations where reduced energy expenditure is experienced by using a staggered formation [8], [23].

Comparing the two configurations, it is shown that in-line configurations are more susceptible to influences from phase lag and spacing between foils, offering higher potential thrusts but also increased risk of destructive interference [7], [9], [11]. Staggered reveals to yield more modest thrust gains on the usual but on the contrast, offers higher robustness and stability, observed to be more significant at higher Reynold numbers [13]. This observed trade-off is paramount to biological and engineering applications, providing crucial context for this study.

This project is aimed to focus on in-line configuration of tandem foils. This is because it provides more straightforward opportunities to capitalise on wake-foil interactions and explore the effects that foil geometry has on thrust and efficiency.

2.2.2 Spacing and Phase Lag

As abovementioned, the performance of tandem flapping foils, particularly in in-line configuration, strongly depend on two parameters, namely spacing ratio (S/c) and phase lag (φ). Spacing ratio refers to the distance between the leading and trailing foils along the flow, normalised by the chord length. Phase lag is what defines the temporal offset between the two foils in oscillatory motion.

According to a study done by Muscutt et al. [7], it is shown through experiments that the trailing foil's performance is closely related to the spacing ratio. When the spacing ratio is relatively close – ($S/c < 2$), the trailing foil operates more unpredictably with the upstream wake's strong vortical structures. This leads to two ends of outcome, either increasing thrust or degrades performance, depending on the timing. At moderate, optimal ranges of $S/c \approx 2 - 4$, the wake interference is now constructive rather than destructive which can boost the efficiency of thrust and propulsion, achieving gains reaching 50% in thrust over a baseline single foil. On the other spectrum, when spacing ratio is large ($S/c > 5$), wake effects become negative due to coherence decay. The benefits then begin to diminish which causes the trailing foil to function very similarly to an isolated system [7], [9]. Another study, through simulations, numerically confirm the above observations, showing

that past an optimal range, efficiency and performance decreases drastically with increasing spacing ratio, or distance between the foils [10].

With regards to phase lag (φ), the timing of the oscillation of the trailing foil has a direct impact on the generated thrust and efficiency. When the trailing foil is tuned to phase in a way that its oscillation relates well to the vortical structures, flow around the airfoil will be improved which leads to better thrust generation. It has been studied that this improvement of thrust generation can go up to 25% when the phase lag occurs at close to a quarter cycle [11]. Furthermore, when poorly tuned, the phase lag can cause negative, destructive interferences between the foils' vortices, leading to fluctuations in the thrust produced and reduced efficiency [9].

Phase lag has been linked to reduced frequency (k) and Strouhal number (St) to be able to achieve its optimal timing [12]. In addition, the system's efficiency is best when the trailing foil's movements and the leading foil's vortex shedding in the upstream wake are synchronised [10]. A study by Joshi and Moya was conducted and confirmed that the wake-induced flow dynamics are influenced by both spacing and phase lag [27]. They also identified scenarios where constructive wake capture improves efficiency significantly but also where misalignment can cause a reduction in said efficiency and performance.

To investigate the influence of both spacing and phase lag together, several studies were conducted, and they emphasised on the interdependency of both factors instead of their individual, isolated influence. It was discovered that the best case scenario often involved moderate ratios of spacing alongside appropriately tuned phase lags [10] while the thrust produced usually drops when either of the parameters deviate from the optimal values [7]. As studied by Joshi and Moya, the effects caused by wake-foil interaction can vary based on the changes in values of either phase lag or spacing [27]. This highlights the significance of these parameters to tandem foil effectiveness.

Relating back to this study, since it is a palm grass foil configuration that is being studied, wake structures that occur are likely to differ from that of rigid foils due to differing geometry. The differing geometry may cause the optimal ranges of phase lag and spacing ratio to shift.

2.2.3 Synchronisation and Wake Control

In tandem foil configurations, the synchronisation of the foils play an important role in the resulting wake dynamics and therefore, propulsion. Drawing from biological examples, the synchronisation among schools of fish or flocks of birds allows each individual to take advantage of the unsteady flow generated by their adjacent counterparts. This allows them to minimise the energy used when maintaining the cohesiveness of the group [8], [23]. Translating this into engineering applications, the tandem configurations systems can be adjusted in a way that synchronises the trailing foil with the vortical structures from the leading one, thus utilising the flow to draw energy and generate thrust.

Synchronisation of the foils in their motion allows for wake control to be achievable. It is shown that through synchronisation of the trailing foil with the upstream vortex shedding, constructive wake control can be achieved, leading to coherent vortices

that reinforce flow circulation [7], [11]. On the contrary, failing to synchronise can lead to the disruption of the wake and therefore poor performance due to unsteadiness [9], [27]. A study by Akhtar et al. [10] emphasises that with proper active synchronisation of the foils and their movements, it allows for wake interactions to be better adjusted over a larger range of flow conditions which affords improved control than passive configurations.

As mentioned, there are two ways in which wake control can be attained – passively or actively. Passive configurations refer to when the trailing foil's oscillation alignment to the unsteady wake from the upstream foil is natural. Active refers to purposeful adjustments and alterations to frequency, spacing, phase lag, etc. [10], [12]. In nature, biological creatures tend to adopt passive synchronisation as the downstream individual intuitively adjust to the wake generated from the leader or adjacent members [8]. However, in engineering systems, it is much preferable to have active synchronisation as it offers more flexibility to vary the tandem foil performance based on the parameters such as flow velocity, Reynolds number, and other factors suitable for the experimental objectives [10], [12], [27].

Over time, research has shown that tandem foils in synchronisation tend to have convincing improvements in thrust gains and propulsive efficiency than isolated foils. An observation of an increase of up to 25% in thrust was seen when the trailing foil synchronises with desirable vortex structures in a study done by Sampath et al. [11]. Meanwhile, it was also identified that the presence of synchronisation can reduce negative interferences and encourages stable wake patterns [27]. In addition, being able to synchronise the foils can assist in a phenomenon called wake steering, allowing the tandem foil configuration to perform better in propulsion or manoeuvring [7], [9].

Linking back to palm grass foils, it is important to understand how synchronisation might affect the thrust and propulsion produced where necessary. This is because synchronisation may have slightly different results on palm grass foils due to its varying geometry from conventional tandem foils.

2.3 Foil-Wake Interaction

2.3.1 Wake Structures and Vortex

The vortex structures that are formed in the wakes of the foils are what essentially determine the performance of the system. The main components are the Leading Edge Vortex (LEV) and Trailing Edge Vortex (TEV) which ultimately determines the thrust, efficiency and stability of the system through their strength and coherence.

In the process of heave and pitch cycle, instantaneous lift is affected by an LEV that forms on the suction side of the foil. Suppose the LEV is shed appropriately, this LEV will become a contributing factor in generating thrust through the enhancement of circulation [4], [6], [16]. The downstream wake is established through the complementary action of the TEV by controlling the periodic shedding. The synchronisation of these two processes is crucial to develop stable thrust generation.

There are multiple wake types that can be formed from the foils depending on a variety of factors such as Strouhal number, reduced frequency and other kinematics [23].

1. **Von Kármán:** this refers to a drag producing wake that occurs due to opposite sign, alternating vortices.
2. **Reverse Von Kármán:** this is the thrust producing wake that is caused by momentum being directed downstream because of positive LEV-TEV interactions at the optimal ranges of Strouhal number.
3. **Deflected jet wake:** this is a manoeuvre related wake due to asymmetry in the strength or shedding of vortices [9], [27].

For tandem foil configurations, appropriate LEV-TEV alignment creates the possibility of the trailing foil capturing the circulations produced and therefore, increase the thrust output. On the other hand, misalignment leads to interferences that destroy and destabilise the system, dropping the efficiency [7], [10], [11], [27]. As aforementioned, the study conducted by Joshi and Moya [27] revealed that the combination of both spacing and phase lag significantly affected the wake dynamics. This produces situations where vortex behaviour can either enhance or diminish the thrust gained.

As observed in nature where similar individuals take advantage of vortex mechanisms to minimise energy expenditure, this concept can also be applied to engineering [8], [23]. Being able to use accurate positioning, timing and synchronisation will be able to improve engineering performance.

2.3.2 Wake Capture Mechanism

This mechanism refers to the ability of the trailing foil to elicit useful energy from the unsteady flow from the leading foil. In flapping foils, this is usually in the form of trailing foil interacting with the LEV and TEV from the leading one. There are two types of wake capture that will be covered – passive and active.

In passive wake capture, it tends to be observed more abundantly in nature. The exploitation of upstream vortices in schools of fish or flocks of birds can significantly reduce the energy spent by each [8], [23]. In a study of flexible foils, it was observed through experiments that when placed in vortex wakes, they can encounter passive propulsion [28]. This meant that the foils were able to extract energy from the vortices without any active control.

For active wake capture, it is more applicable to engineering systems such as the tandem foil system. These often rely on active control to capitalise on the upstream vortices. In earlier studies, it was shown how the adjustment of frequency and phase lag – of the downstream foil in particular – grants synchronisation with the vortices from the upstream which eventually leads to increased thrust [10]. Additionally, it is also verified that active control and tuning in spacing and phase lag can augment negative interferences from the vortex into beneficial interactions that enhance performance, coherence and output [27].

The essence of wake capture is not exclusive to propulsion but can also be used in the extraction and harvesting of energy. It was demonstrated that the harvesting of

energy can be done via the oscillations of flapping foils that convert vortex motions into mechanical output [29]. It was determined that the frequency and amplitude of the oscillations relative to the incoming flow must be tuned well to achieve the best power output. In fact, it may be the case to best capitalise the tandem foil kinematics is to intentionally exploit upstream wakes to in turn, augment the harvested energy [15].

Vortex foil phasing, a principle that drives wake capture refers to when the vortex that arrives at the trailing foil is in phase with the cycle. The induced velocity is then added constructively as a result of that to the motion of the foil, therefore increasing the effective angle of attack as well as circulation [5], [12], [27]. However, should there be a mismatch in timing, destructive interference may occur which leads to decreased thrust [9], [7]. In a study done by Alaminos-Quesada and Fernández-Feria [12], a framework was provided to predict when wake capture would improve performance by using linear potential theory.

2.3.3 Theoretical Models

Many theoretical and computational models have been developed over time to predict both aerodynamic and hydrodynamic performances of flapping foils in unsteady flow. They aim to capture the relationship between kinematics, wake dynamics and the thrust and efficiency that is a result of the former, all while reducing the need for costly experimental trials.

Linear potential flow theory was used in early studies to get an approximation of unsteady lift and thrust generated on the oscillating foils. They usually encompass the assumptions of inviscid, incompressible flow along with small amplitude motions that allow for closed form expressions of forces and power coefficients. This theory was applied to flapping foils in tandem and led to the derivation of formulations that accounts for the interaction between heaving and pitching of both foils [12]. The model applied in this study was successful in predicting thrust and efficiency trends that align with experimental observations done at moderate Reynolds numbers. In addition, Theodore Theodorsen's unsteady aerodynamic theory gives the foundational expression for oscillating foils, which combines circulation and wake behaviour through a transfer function dependent of frequency.

2.4 Effect of Foil Geometry

2.4.1 Effect of Thickness and Camber

The geometry of the foil has a significant influence on its characteristics and performance in unsteady flow. The two most important geometrical features are the thickness ratio and camber, both of which affect pressure distribution, behaviour at the boundary layer, as well as the wake structure of the foils. When varied, they affect and alter the stability of LEVs, influence flow separation and as a result, affect thrust and efficiency.

Thickness

To address thickness ratio, it is what determines the hydrodynamic loading and the wake characteristics of the flapping foils. In a study done by Lagopoulos et al. [9], experiments were conducted by using hydrofoils of varying thickness. It was

Ratio

highlighted that foils that were thicker had the tendency to generate stronger yet less coherent vortices, which resulted in a jet wake that was either deflected or broadened. This was linked to flow separation occurring earlier, closer to the leading edge of the foil as well as increased form drag. On the contrary, thinner foils was observed to have cleaner vortex shedding. Though perhaps weaker in strength, the produced reverse von Kármán wake was more stable, which allows for better thrust generation.

Similarly, another study by K. Isogai et al. [30] observed that an increase in thickness of the foils amplify dynamic stall effects, which may in turn improve instantaneous lift but reduce the mean efficiency in propulsion. It was also numerically confirmed by Akhtar et al. [10] that slight changes in a foil's thickness ratio can alter the optimal Strouhal number (St) and reduced frequency (k) required to maximise thrust. In more recent studies and simulations, it has been seen that varying thickness also plays a role in wake symmetry and in the case of tandem configurations, inter-foil coupling [27]. This affects both positive and negative interference regimes.

Linking thickness ratio to energy harvesting, it has been highlighted that finding the optimal foil thickness can help balance both structural integrity with hydrodynamic response. Excessive thickness has been observed to limit the amplitude of oscillations, while too thin of a foil leads to the foil being prone to early separation and instability. Hence, finding the right thickness ratio is such a critical factor in balancing thrust and efficiency alongside stability.

Camber

The second crucial parameter is the camber which introduces asymmetry to the geometry of the foil. This asymmetry leads to a modification of the foil's effective angle of attack and altered lift generation. In the case of flapping foils, at lower heaving and pitching amplitudes, cambered foils tend to produce greater lift and thrust as a result of desirable pressure gradients on the suction or low pressure side of the foil [30], [31].

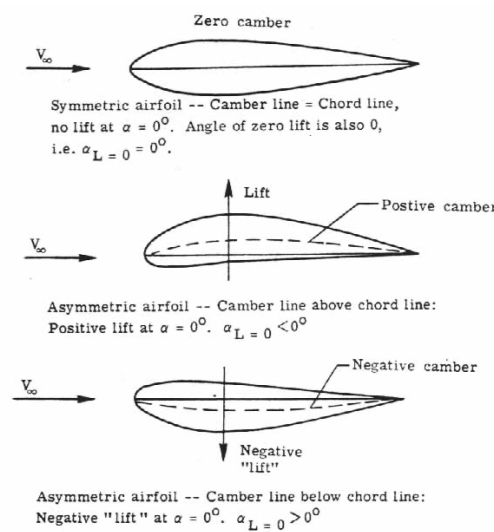


Figure 4 Cambered vs Non-Cambered Airfoil [32]

Excessive camber, however, can lead to early flow separation on the foil, which leads to reduced lift and efficiency at higher amplitudes. It has been demonstrated that when a slight, positive camber is introduced to a foil, it can improve the efficiency of energy harvesting when studying oscillating hydrofoils in a power generation regime. This suggests that a foil's camber can be tuned and adjusted to extract optimal performance based on objective [24].

In the case of nature and therefore, bio-inspired applications, swimmers and aerial beings often possess slightly cambered or flexible foils which helps to maintain LEV attachment and suspend stall [4], [8], [23]. This can link to a possibility that bio-inspired designs such as the palm grass foil might present similar advantages of vortex formation stabilisation since it also possesses similar curvatures and non-uniform geometry.

2.4.2 Bio-Inspired Foil Shapes

Over the years, researchers have begun to explore bio-inspired and non-conventional geometry of foils in hopes to mirror the efficiency and adaptability of natural locomotion. Dissimilar to the current conventionally symmetric or rigid foils, nature's swimmers and flyers usually possess geometries that are asymmetrical and dynamically flexible, which allows the manipulation of vortical structures to positively affect propulsion and manoeuvring [8], [23], [4].

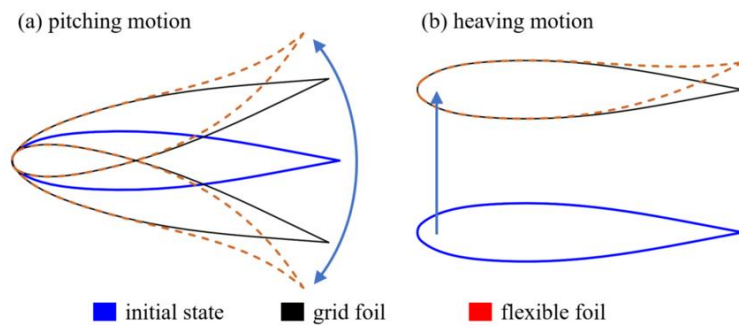


Figure 5 Pitching and Heaving Profile Deformation of Rigid vs Flexible Foil [22]

Inspiration from Biological Foils

Researching natural foils found in fish and birds such as fins and wings, Triantafyllou et al. [4] and Taylor et al. [23] found that they operate at Strouhal numbers that offer high efficiency within a range of $0.25 \leq St \leq 0.35$, while depending on surfaces that are capable of passively adjusting based on the flow condition. This flexibility that is observed allows for geometric variations that ultimately grant dynamic control of the LEV whilst preserving stability and delay of stall. In two separate studies, it was observed by Beat et al. [28] and Zhu [5] that these flexible foils provide a structure that can capitalise on unsteady flow when it comes to propulsion and energy harvesting.

Drawing from this, more experiments and research has been conducted to examine the relationship between non-conventional foil geometry (e.g. trapezoidal, elliptical, non-uniform, etc.) and formation of wake alongside generation of thrust. A study on deflected wake interactions in tandem configuration by Lagopoulos et al. [9] found that geometry with asymmetry or taper was able to encourage flow redirection and vortex steering. Separately, it was also noted that variations in thickness distribution

can influence coupling between foils and alter the effects of wake interference between constructive and destructive [27].

Additionally, configurations that involved multiple and flexible foils were explored and the results showed that non-uniform foil geometries were able to improve stability as well as curb unsteady loading better as compared to the more conventional rigid and symmetric foils [10], [13].

Palm Grass Foil

Aside from fishes and birds, grass blades and leaves also provide inspiration for the palm grass foil due to their exhibition of thickness variation and natural curvature. These allow for flexibility and vortex alignment in the presence of unsteady flow, which parallels flexible fins found in fish and their hydrodynamic responses [4], [8]. This non-uniformity found in the geometry of the palm grass foil is predicted to augment both LEV and TEV, and through flow adaption, increase the potential of wake stabilisation and thrust enhancement.

These bio-inspired designs point to engineering opportunities such as developing quieter, higher efficiency propulsion systems and energy harvesters that can adapt to flow, potentially surpassing the conventional rigid symmetrical configurations.

2.4.3 Leading and Trailing Edge

Vortex formation, pressure distribution and wake stability are factors that determine the performance of flapping foils and the thrust generated by them. These are crucially influenced by the geometry of both the leading and trailing edge of the foils in flapping motion. Minute variations in the geometry like curvature, thickness or flexibility can dictate if flow remains attached or goes through premature separation.

The leading edge (LE) is the point where the LEV – a contributing factor to lift and thrust in unsteady flow – is initiated. If the LE is sharp, stronger instantaneous lift may be generated due to an early vortex roll-up. On the other hand, an LE that is rounded or more cambered can aid in delaying flow separation from the foil, this maintaining attachment of the LEV for longer periods [16], [30]. It has also been found that moderate curvature in the LE leads to pressure gradients that are more gradual and therefore, better efficiency through the stabilisation of LEV shedding [31].

A sharp LE can increase the peak thrust at higher amplitudes of oscillation but leads to increased consumption of power and dynamic-stall losses [16], [30]. Contrary to that, foils that bio-inspired designs of flexible or compliant LE promotes partial deformation which aids in mitigating stall and encourages smoother detachment of vortex [4], [8].

The trailing edge (TE) of a foil dictates TEV shedding and the coherence of the downstream wake. It is desirable for the TE to be sharp since it encourages alternating vortex shedding which aids in the formation of stable reverse von Kármán jet [5], [9]. However, when the TE is overly sharp or inflexible TEs can lead to high-frequency oscillations and acoustic noise, both of which are avoided in propulsion.

Similar to the LE applications, flexible TEs have been observed to reduce unsteady forces and fluctuations in drag. In the study by Triantafyllou et al. [4], it was observed that gradual curvature in the TE – as seen in fins of fish – can improve momentum-jet symmetry, therefore resulting in better stability in thrust. Synthetic foils that adopt elastic or feathered trailing edges have also been reported to reap similar benefits [33].

When combined, both LE and TE designs dictate the shedding and merging of vortices. Slight cambers in the LE and tapers in the TE has shown to generate stronger circulation loops of the LEV and TEV, also improving the phase coherence between them which leads to a higher average thrust produced [10]. Similarly, Lagopoulos et al. [9] revealed that asymmetry in the edges can aid in redirecting wake momentum which in turn also augments thrusts.

2.5 Past Experimental Approaches

2.5.1 Water Tunnel

Apart from wind tunnels that often require more space and finances, water tunnels remain as one of the more widely used methods of experimentation and testing of flapping foils in unsteady flow. Water tunnels provide the controlled environment required to measure time-based forces, flow fields as well as wake dynamics, particularly at more moderate values of Reynolds numbers that are more relatable and applicable to bio-propulsion and smaller-scale engineering applications [6], [7], [9].

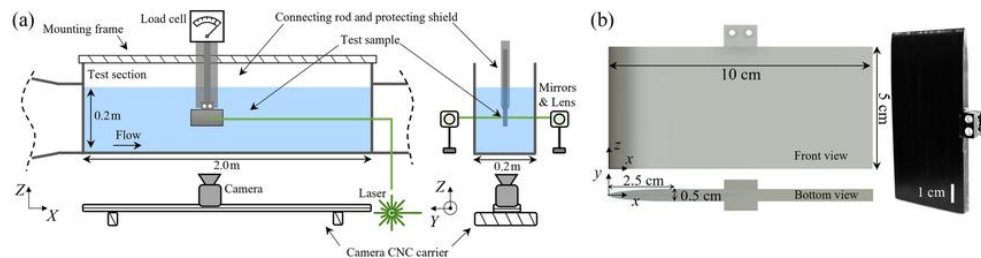


Figure 6 Example of a Water Tunnel Schematic (a) Front View (b) Side View [34]

In a typical experimental setup, a recirculating water tunnel, flapping foils, sensors and flow-visualisation equipment (where necessary) such as injection of dye or particle image velocimetry (PIV) are involved. The foil is then mounted onto a support rig or actuation system that imposes heaving and pitching motions about an axis or pivot point. Commonly, data are recorded for a range of Strouhal numbers (St) and reduced frequencies (k) to then quantify the trends of lift, thrust and efficiency [5], [16], [30].

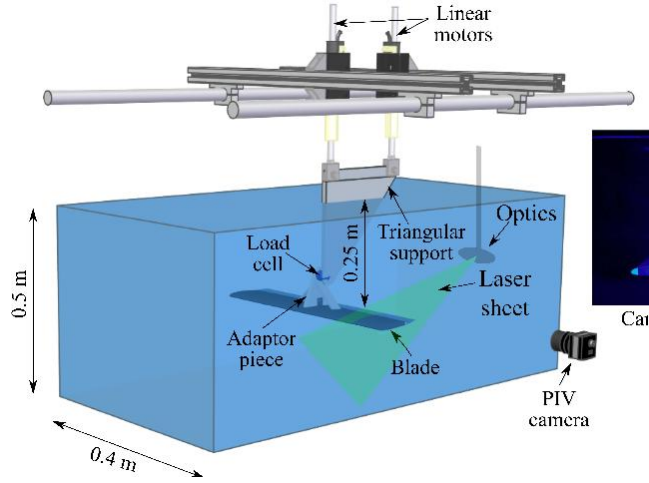


Figure 7 Example Schematic of Water Tunnel Setup with Flapping Foil [35]

In a study by Muscutt et.al [7], tandem foil experiments were conducted using a closed-loop, recirculating water tunnel to determine the relationship between spacing and phase lag with thrust. In their setup, they captured instantaneous hydrodynamic forces with phase-locked synchronisation by utilising load cells of high resolution. A separate study by Lagopoulos et al. [9] further improved the former setup by introducing PIV and high-speed imaging to capture vortex structure visualisations which allowed the study of correlation between forces and wake topology to be possible.

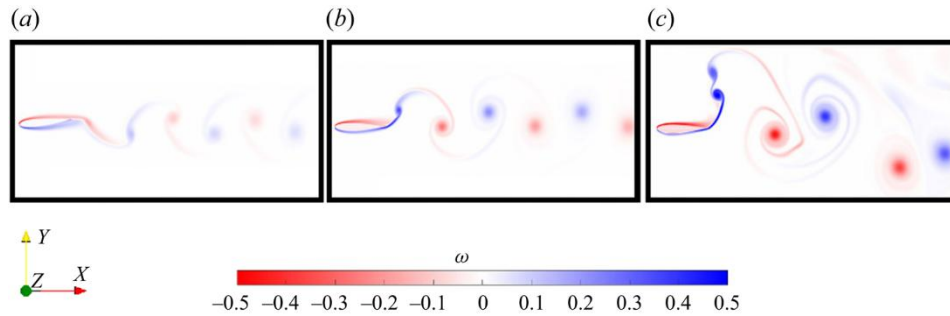


Figure 8 Example of Wake Visualisation at $St_C=0.625$: (a) von Kármán Street at $A_C = 0.2$; (b) reversed von Kármán street at $A_C = 0.3$; and (c) deflected jet at $A_C = 0.4$ [9]

In such experiments, force and torque are usually measured through 6-axis load cells or strain gauges that are mounted at the support rigs or areas surrounding the foil. Rotary encoders or laser sensors that detect displacements are used to capture position and motion data, such as heaving and pitching. Lastly, the synchronisation between the control of motion and data acquisition allows for more accurate averaging of oscillatory cycles [10], [27].

As for flow visualisation, PIV or laser-induced fluorescence (LIF) are commonly used to reconstruct wake fields. It was demonstrated in the study by Godoy-Diana et al. [33] that PIV can aid in resolving transitions between periodic and deflected wake regimes, which in turn reveal the sensitivity of the jet orientation to the pitching and heaving amplitude. In other energy harvesting studies and experiments, additional equipment like torque transducer sensors or voltage pickup sensors may be employed to measure other data such as efficiency of power extraction [5], [29].

Some common non-dimensional parameters that are investigated in such water tunnel experiments and studies are:

- Strouhal number (St) which governs wake pattern and thrust efficiency.
- Reduced frequency (k) which relates the oscillation rate and flow speed.
- Amplitude ratio (A^*/C) which defines motion relative to the chord length.
- Reynolds number (Re) which determines effects of viscosity and flow regime.

Any variations in any of these parameters can systematically allow the transformation of drag producing von Kármán wakes to those that produce thrust instead [16], [30], [23].

Our palm grass foil study canonically follows a similar framework for experiments. The 3D-printed foils that are of varying geometry – shape and thickness – are put into a water tunnel for testing. The foils then oscillate through a controlled axis in pitching and heaving, and the displacement data is collected using sensors at a fixed point of the support rig. The measurements captured are then converted from time-domain into frequency domain.

2.5.2 Flow and Wake Measurement

Accurate visualisations and measurements of wake and vortex structures in an unsteady flow can help in understanding the wake dynamics of flapping foils. Over the years, there have been many advancements and improvements in experimental diagnostics which have assisted researchers to capture detailed information such as vortex structures, wake topology and distribution of momentum. Having these visualisations provide the premise of linking flow patterns observed to propulsive performance [7], [9], [10]. These techniques can be branched into two types, qualitative and quantitative.

Qualitative Techniques

In earlier research and studies, they primarily relied on techniques like dye injection, hydrogen bubble methods or smoke visualisation to qualitatively reveal the formation and evolution of vortex and wake. In the study by Lagopoulos et al. [9], the deflected jet produced by a lone foil was visualised followed by a trailing foil could help in the correction of the jet. This was done using time-resolved images to link the angle change of the jet to the thrust produced. Dye methods are valuable for quick, low-cost visualisations of wake formation – whether it is von Kármán or reverse von Kármán wake – jet deflection, wake symmetry and more. These are primarily used in preliminary experiments.

(Digital) Particle Image Velocimetry (DPIV / PIV)

PIV remains as a go-to method and tool in experiments that deal with foils, vortices and wakes because they offer many advantages. These advantages include phase-averaged velocity and vorticity fields; it gives the possibility to integrate momentum and therefore infer thrust; lastly, it provides clear and direct visualisation of LEV, TEV and their timings. These are all crucial aspects of investigating tandem foils and spacing.

An experimental study by Anderson et al. [6] used DPIV in their water tunnel investigation to measure the velocity fields that arise from the wake produced by the oscillating foil in both heaving and pitching. The DPIV gave conclusive observations

on how reverse von Kármán jets that point toward the downstream direction are heavily linked to higher efficiency. This was later backed up by Godoy-Diana [33] who similarly diagnosed the same theory after mapping the transition from von Kármán to reverse von Kármán wakes and then subsequently to asymmetric von Kármán wake at higher amplitudes whilst using 2D PIV.

Separate studies by Muscutt et al. [7] and Joshi and Moya [27] also used PIV or tools similar to PIV to observe the relationship between the trailing foil with oncoming vortex. This allowed explanations for why there are optimal spacings and phases that can give rise to an increase of thrust of about 50% while other spacing values can lead to loss of thrust generated.

Flapping foils usually operate on a strongly periodic basis and flow, as such, many utilise phase-locked or time-resolved PIVs that are synchronised with the motion controller and force balance. An example of this lies in the study by Lagopoulos et al. [9] where they were able to link the timing when the foil in the downstream essentially “pulls in” a deflected jet to the rise or decrease in thrust. Despite its relative cost being much higher, time-resolved PIVs are usually the clearest way available to bring out observations such as vortex pairing, cancellation or capture in tandem configurations.

Quantitative Techniques

Some quantitative techniques that are used are laser doppler velocimetry (LDV) as well as Schlieren or shadowgraph. LDV and hot-film or pressure taps are used concurrently with PIV as support, producing high frequency, single point velocity or surface pressure data. These data are impactful to studies as they detect flow separation or dynamic stall cycles [30]. Schlieren and shadowgraphs are more common in air-based flapping foil studies and are less relevant to this study.

2.5.3 Numerical Simulations

Having numerical simulations can complement experimental studies very well, providing a way to achieve more detailed insight and information into observations made of flapping foils in unsteady flow, their pressure distributions and their energy conversion mechanisms. Numerical simulations have been made indispensable for exploring parameters, especially with advancements in computational and digital resources as well as solver accuracy, allowing the optimising of geometries and confirming theoretical hypotheses [10], [12], [13], [27].

When it comes to computational strategies, many flapping foil simulations involve solving the Navier-Stokes equations of unsteadiness and incompressibility, using both finite-volume and finite-element methods. 2D simulations of tandem configured foils in oscillation were conducted by Akhtar et al. [10] to resolve both pressure and velocity fields, therefore showing strong wake pattern correlations. Computational Fluid Dynamics (CFD) simulations are also a common tool used to complement experimental studies. A CFD code was used in a study by Lagopoulos et al. [9] to model deflected-wake interactions and hence, validating that like experimental observations, wake redirection and augmentation of thrust can be calculated numerically. High resolution CFD was utilised by Joshi and Moya [27] at Reynolds numbers ranging between $10^3 - 10^4$ to study the relationship that wake

coupling has with spacing ratio and chord ratio, therefore accurately replicating thrust trends observed in experimental studies.

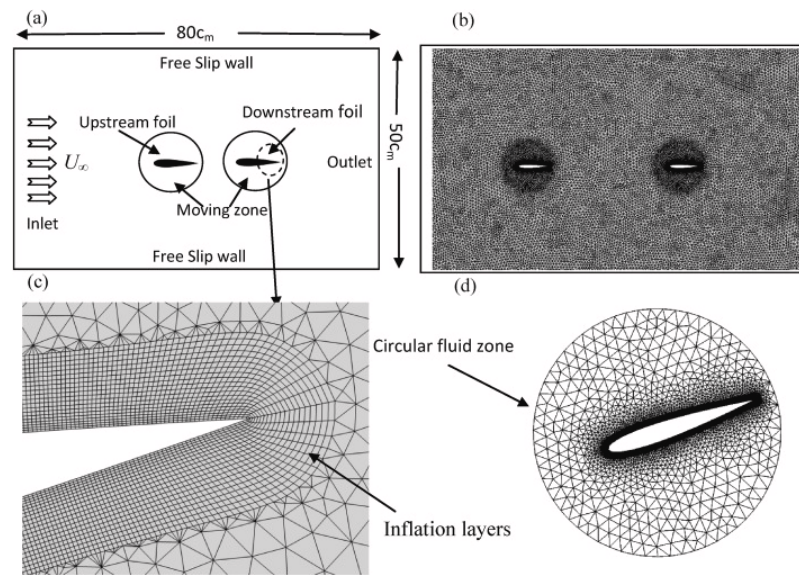


Figure 9 CFD images of tandem flapping foils: (a) The upstream and downstream foil location (not to scale). (b) Meshed surface of inner and outer fluid zone. (c) Closer view of the meshed aerofoil tip (d) Meshed circular fluid zone. [36]

Other methods include the immersed boundary method (IBM) and the fluid-structure interaction (FSI) method. IBM has been deemed effective when it comes to problems that involve boundaries that move within a fixed grid. IBM was used in a study by Kumari and Chakraborty [13] to simulate multiple flapping foils operating under free surface conditions and confirm surface wave pattern observations in dye visualisations. FSI is used more commonly together with CFD solvers with structural dynamics in flexible and bio-inspired foils as their frameworks are able to capture the effects of deformation on thrust and efficiency [31]. These methods have shown that with the presence of moderate structural compliance, there can be enhancements to the wake symmetry and overall propulsive efficiency, which aligns with biological examples [4], [8].

A cheaper alternative of investigating includes discrete vortex methods (DVM) and linear potential flow (LPF) formulations. Potential theory was used in a study by Alaminos-Quesada and Fernández-Feria [12] to replicate tandem foil interference in order to predict improvement of thrust for certain phase lags. Though these methods tend to neglect viscous and separation effects, they usually enable quick sweeps on the parameters necessary.

When determining whether numerical simulations align with experimental observations, validations need to be carried out. This is usually done by comparing factors and parameters such as thrust coefficient (C_T), efficiency (η), and wake topology. Reference datasets for single and tandem foils can be established and crosschecked against in subsequent CFD or other numerical solutions, providing a benchmark to compare against [6], [7]. In the studies by Lagopoulos et al. [9] and Joshi and Mysa [27], their numerical predictions came close to their experimental results which confirms that when the CFD is properly resolved and formulated, it is

able to capture LEV/TEV formations, vortex coupling and phase-dependent thrust variations within a 5% margin.

In the case of flexible or non-symmetrical foils, it has been shown that through CFD like computed pressure distributions, predictions for stall delay and thrust enhancements that arise from moderate camber can be done accurately [31]. With such alignment between both numerical simulations and experimental observations, it establishes present-day efforts to maximise the use of computational tools for the optimisation of foil geometry and design in the area of propulsion and energy harvesting.

3. METHODOLOGY

3.1 Problem Definition

In this study, the objective is to investigate the relationship between the foils' geometry – shape and geometry – and how it affects the performance of the foil in energy harvesting applications. By using multiple sets of foils of varying thickness and shape, the study aims to identify the ideal foil geometry configuration that yields optimal values of efficiency, thrust and power output under controlled flow conditions.

3.1.1 Foil Geometry

The test model foils were all 3D-printed using polylactic acid (PLA) and mounted as rigid foils. A baseline pair of foils is 3D printed with a flat shape and small thickness. (insert picture). Following that, the foils are grouped into sets, with each set consisting of foils of the same planform shape but varying thickness. The sets are split into two categories:

1. Shaped Edges – Flat Mid-Section
 - Only the edges – Leading and trailing edge – follow a geometric profile
 - The mid-section of the foil remains flat
 - Visually, they resemble a rectangular foil with modified edges
 - There are 6 sets in this category
2. Fully Shaped Foils
 - These follow the designated shape for the entire planform
 - Both the edges and mid-section are non-rectangular and flat
 - There are 5 sets in this category

This structure and categorisation enable a controlled study on how both edge shape variation (Category 1) and fully non-conventional planforms (Category 2) can affect energy harvesting applications.

3.1.2 Problem Statement

Within these constraints of varying geometry of the foils, the aim is to investigate and quantify how the geometry of the foil affects measuring parameters such as Strouhal number, force, power and efficiency. This is then translated to determine the optimal foil geometry for energy harvesting applications.

3.2 Evaluation

When determining the ideal foil geometry for energy harvesting applications, the performance of each 3D-printed foil is evaluated based on a combination of parameters. These parameters are derived from experimental measurements of the oscillatory motion of the foils and are then compared across each set of foil shape and thickness.

The primary data collected is based on experimental data that is obtained through water tunnel testing alongside positioned sensors that detect the displacement of the foil over time (displacement-time). This displacement-time data is processed post-experiment

and is used to infer hydrodynamic quantities through kinematic measurements, spectral analysis and damping-based force estimation methods revised from other flapping and tandem foil investigation researches [6], [9], [7], [27]. This information is then converted into the frequency domain to identify the domain frequency and period. The following subsections highlight the main parameters investigated and how they affect the results.

3.2.1 Strouhal Number

Reduced frequency and Strouhal number are both non-dimensional parameters that assist in describing the kinematics of flapping foils. They play a significant role in characterising wake regimes as well as interactions between foil and flow.

The reduced frequency represents the ratio between the foil's oscillation speed against the speed of the flow relative to the foil. It can be defined as:

$$k = \frac{fc}{U_{\infty}} \quad (11)$$

Where:

f = frequency obtained via Fast Fourier Transform (FFT)

c = chord length

U_{∞} = freestream velocity

The Strouhal number as defined earlier:

$$St = \frac{kA}{\pi c} = \frac{fA}{U_{\infty}} \quad (12)$$

Where:

A = peak-to-peak amplitude

This parameter is one that highly correlates to the performance of the foil in propulsive and energy harvesting applications. It has also been observed in biology that the ideal range of Strouhal number lies within the range of $0.2 < St < 0.4$, allowing the maximisation of jet coherence and efficiency in energy transfer [23], [4], [5].

3.2.2 Force

Within this experiment, instantaneous lift and moment that act upon the foils are inferred using the dynamic response method that has been used in similar studies [9], [27]. In this method, the flapping foils are modelled as damped harmonic oscillators, where the inertial and damping components balance the hydrodynamic forces of the system.

For heaving:

$$F_L = m\ddot{h} + c_h\dot{h} \quad (13)$$

And for pitching:

$$M_\theta = I_\theta \ddot{\theta} + c_\theta \dot{\theta} \quad (14)$$

Where:

m = mass

I_θ = rotational inertia

c_h = coefficient for heaving

c_θ = coefficient for pitching

Through free-decay tests, the damping coefficients are determined experimentally [37]. The coefficient of heaving is determined by using an exponentially decaying sinusoid to a free-release displacement test:

$$h_d(t) = A_h e^{-\frac{c_h}{2m}t} \sin(\omega t + \phi) \quad (15)$$

While the coefficient of pitching is determined through a pitch-release test while using the over-damped exponential form:

$$\dot{\theta}(t) = A_\theta e^{-\frac{c_\theta}{I_\theta}t} \quad (16)$$

The phase-averaged coefficients are then defined as the following:

$$C_L = \frac{F_L}{\frac{1}{2}\rho U_\infty^2 b c} \quad (17)$$

$$C_M = \frac{M_\theta}{\frac{1}{2}\rho U_\infty^2 b c^2} \quad (18)$$

Where:

b = span of the foil

ρ = density of the fluid

In the case of tandem configuration analysis, the mean aerodynamic coefficients are expressed as the averages of the fore (F) and hind (H) foils where necessary, resulting in a slightly altered equation as seen below:

$$C_{L_A} = \frac{C_{L,F} + C_{L,H}}{2} \quad (19)$$

$$C_{M_A} = \frac{C_{M,F} + C_{M,H}}{2} \quad (20)$$

This convention allows for the comparison of performance between single and tandem foils [7], [9].

3.2.3 Power

Depending on the contributions from both heaving and pitching, the instantaneous power, $P(t)$, can be extracted or imparted by the foil. This can be demonstrated through the following equation:

$$P(t) = F_L(t)\dot{h}(t) + M_\theta(t)\dot{\theta}(t) \quad (21)$$

The equation for the cycle-averaged power can be described by:

$$\bar{P} = \frac{1}{T} \int_0^T P(t) dt \quad (22)$$

Where T refers to a single oscillation period. The corresponding coefficient of power can be then defined as the following for normal and tandem foil configurations respectively:

$$C_P = \frac{\bar{P}}{\frac{1}{2} \rho U_\infty^3 bc} \quad (23)$$

$$C_{P_{Total}} = \frac{\bar{P}_F + \bar{P}_H}{\frac{1}{2} \rho U_\infty^3 bc} \quad (24)$$

This allows for the direct comparison across the different foils comprising of varying geometries and operating conditions. A positive value of C_P indicates a propulsive operation where power is input into the flow. On the other hand, a negative C_P value represents a net energy extraction, henceforth an energy harvesting operation [5], [27], [29].

3.2.4 Efficiency

Efficiency (η) in energy harvesting demonstrates the flapping foils' effectiveness in converting kinetic energy into useful power as output.

$$\eta = \frac{\bar{P}}{\frac{1}{2} \rho U_\infty^3 bA} \quad (25)$$

Where A refers to the peak-to-peak amplitude of the heaving motion. In tandem configurations, the total efficiency can be defined as:

$$\eta_T = \frac{\bar{P}_F + \bar{P}_H}{\frac{1}{2} \rho U_\infty^3 bA} \quad (26)$$

This convention is adopted from several studies, providing an avenue for the efficiency of foils of varying factors such as geometry, spacing ratio and motion amplitude, to be compared more effectively. A higher value of efficiency indicates a more effective extraction of the energy which also corresponds to a stable reverse von Kármán wake as well as appropriately timed interactions between foil and vortex.

Additionally, this experiment follows the Betz limit on efficiency. The Betz limit refers to a theoretical maximum efficiency that a wind turbine can capture, which is said to be at 59.3%. This value is also used as a limit for the efficiency values extracted from the experiments to ensure that the results are coherent and realistic.

3.2.5 Data Processing

Using displacement sensors, the flapping foils' displacements are captured over time. In post experiment data processing, the time-displacement domain data is

converted using FFT analysis into a frequency domain data. Here, the domain frequency and domain period are extracted.

The summary of steps for the data processing is as follows:

1. Raw displacement values $h(t)$ and $\theta(t)$ are recorded in time domain
2. The data is cleaned up, reducing noise and offsets, easing the process for data conversion
3. Using MATLAB, the time-domain values are converted using FFT into frequency-domain values
4. Further analysis is done to compute velocity, displacement, sweep, etc. to evaluate and estimate force (lift) and power
5. These values are integrated over several periods to determine power and efficiency
6. Through normalisation against the constant factors such as ρ, U_∞, b , and c , cross foil analysis can be done

3.3 Experimental Set-up

This experimental study was conducted in the premises of Nanyang Technological University's Water Tunnel facility. In this facility, a controlled environment was ensured using a recirculating water tunnel alongside a custom support rig and sensors. This section of the report details the experimental setup from water tunnel to foil fabrication to mounting and workflow.

3.3.1 Water Tunnel

To investigate the optimal foil geometry for energy harvesting applications, a recirculating water tunnel with a dimension of $0.45m \times 0.55m \times 1.0m$ was used for the physical experiments. The water tunnel comes equipped with an axial pump along with a speed controller to control the flow velocity of the water. The maximum achievable velocity of flow is $0.15m/s$ with an inlet turbulence intensity of $\sim 2\%$. The sides of the water tunnel test section are fitted with tempered glass for optical access during experiments. The open top section of the water tunnel is covered by two acrylic plates – one on each side of the support rig – to minimise the effects of free surface.

3.3.2 Foil Setup

The foils used were 3D-printed using Polylactic Acid (PLA). This material was used due to its stability, rigidity and overall suitability for repeated testing under submersion in water. They were 3D-printed with a consistent infill density and surface finish to minimise variations within each set and group.

As aforementioned, the foils are separated into two categories. A breakdown of the two categories and their naming system is as follows.

1. Winglets – Shaped edges, flat middle section
 - 4 sets – rectangular, round, triangular, concave
 - Thickness varying from 5mm to 20mm; 5mm increments

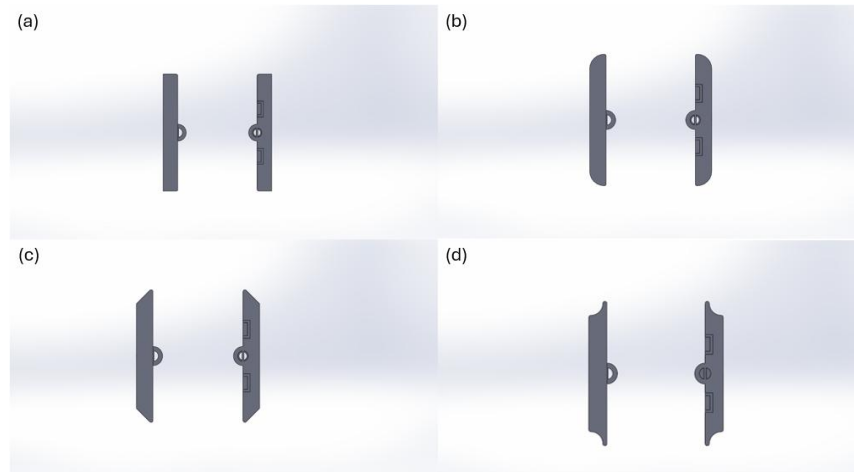


Figure 10 Top View of Winglet Type Foils: (a) Rectangular Winglets, (b) Round Winglets, (c) Concave Winglets, (d) Triangular Winglets

2. Geometric – Fully shaped with no flat areas

- 5 sets – rectangular, elliptical, circular, trapezoid, triangular
- Same thickness variation and increments

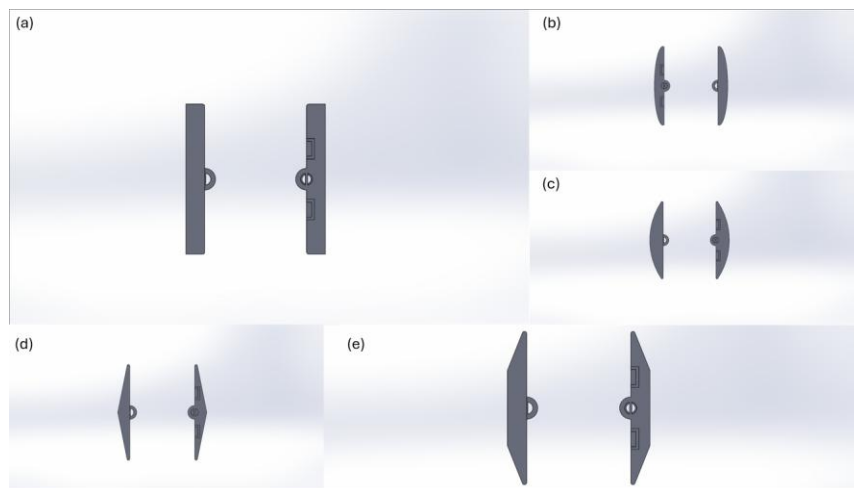


Figure 11 Top View of Geometric Type Foils: (a) Rectangular Geometry, (b) Elliptical Geometry, (c) Circular Geometry, (d) Triangular Geometry, (e) Trapezoid Geometry

In both sets, a reference pair of foils is used as a benchmark for data collection and processing. This pair of foils are thin and flat, having no form of geometry to them. The foils all share the same chord length and total span, c and b respectively, allowing a more consistent comparison across all test models.

The holes in the middle of the foils are meant to allow the connection of the pair of foils through a small steel rod and the slider. The small rectangular slots are meant to be fitted with magnets that will interact with the magnets on the slider.

3.3.3 Mounting of Foils

When mounting the foils, a preparation or assembly segment is first carried out where the foils are connected. As seen, each pair of foils come as two separate pieces

which needs to be connected. With a slider and a short steel rod, along with some bearings, the two foil pieces are connected as seen in Figure 12.



Figure 12 Slider Connection for Foil Pieces

The foils are then mounted vertically onto the custom support rig. The support rig consists of multiple features. Firstly, a large stainless steel rod runs through the rig horizontally. This provides the axis of movement for the foils to oscillate along. At the ends of the rod, also the sides of the support rig, two identical pieces of angled oscillation assisting pieces are mounted. These two pieces aid in nudging the foils to oscillate at the end of each cycle or period. This allows a smooth and quick snapping of the magnets to the oscillating cycle. Next, a cut out at the top of the support rig allows for another rod to pass through. This rod connects the foils to a target piece for the laser sensors to fixate on for the measurement of displacement during the experiment. This cutout is also narrow enough to maintain stability during oscillations without causing too much friction that could hinder the movement of the test models.



Figure 13 Side View of Support Rig for Mounting of Foils

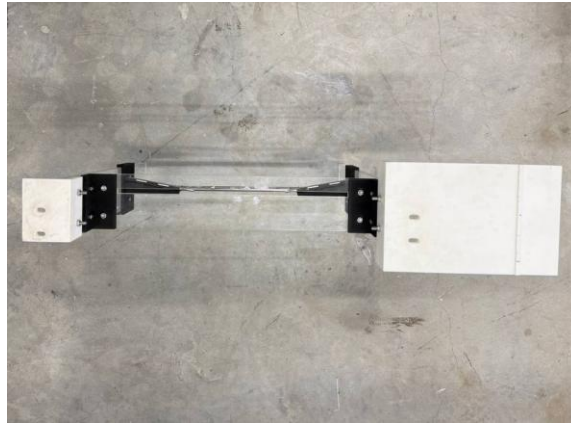


Figure 14 Top View of Support Rig featuring Sensor Mounting Area (Right side) and the Horizontal Cutout

The displacement measuring sensors are mounted on the longer edge of the support rig which provides a consistent placement for measurement. This can be seen in Figure 15.

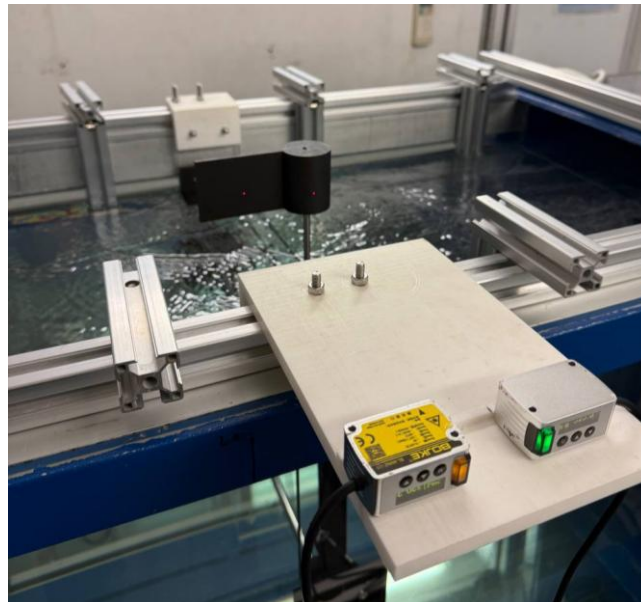


Figure 15 Mounting of Displacement Measuring Sensors on the Support Rig

The foils are then mounted vertically onto the support rig. The sensors are also measured and kept the same distance apart for the experiment day and the data is collected through the black, target piece as also seen in Figure 15. A picture of the foils fully mounted and submerged in the water tunnel can be seen in Figure 16.



Figure 16 Example of Fully Mounted Experiment Setup

In Figures 17 and 18, a hand-drawn schematic diagram of the water tunnel and experimental set up is shown. It depicts the isometric view (Figure 17) and the top view (Figure 18) of the present equipment for the experiments.

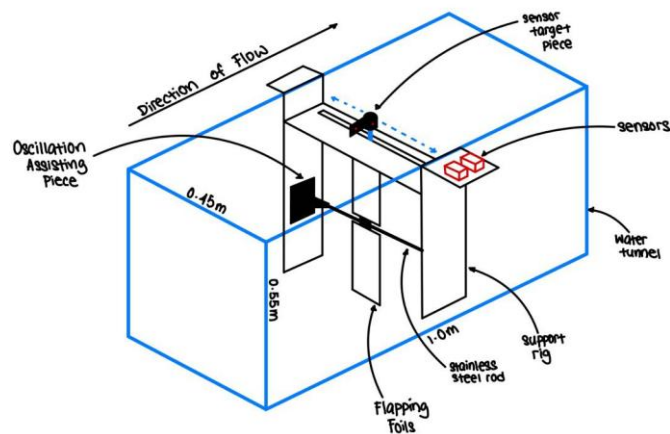


Figure 17 Hand-drawn Schematic of Experimental Set up

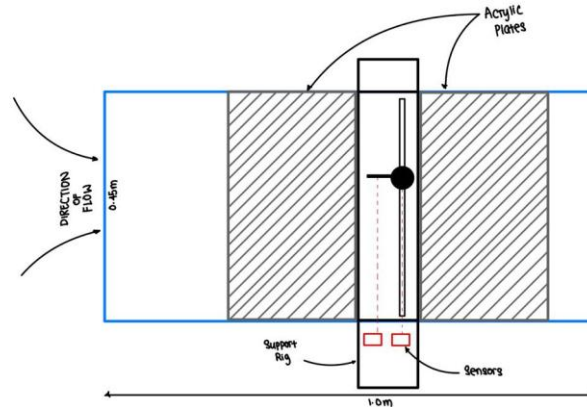


Figure 18 Top View of Schematic Diagram

When mounting the foils, a standard procedure is done to ensure that the set up remains as consistent as possible. Below are the steps to assembling, mounting and sensor placement when mounting the foils:

1. Assembly of foils – inserting ball bearings into the sliders and connecting the foils to make one test piece (see Figure 12)
2. Mount the test piece onto the horizontal steel rod
3. Attach the sensor-target piece through the support rig's cutout
4. Mount the sensors onto the support rig, measuring sufficient and consistent distant between the laser beams ($\sim 50\text{mm}$) – done once per experiment day

3.3.4 Testing Procedure

The testing procedure consists of a series of consistent steps taken across all experiments. Below is an overview of the steps taken for each test:

1. Mount the selected test pair onto the support rig
2. Submerge the experimental set up into the water tunnel
3. Place acrylic plates by each side of the rig to minimise free surface effects
4. Set flow velocity to maximum (0.15m/s) – ensures a consistent factor
5. Allow the foils to oscillate and record displacement over time
6. Repeat Steps 1-5 for all thicknesses of the same geometric shape
7. Data is exported for data processing as seen in Section 3.2.5
8. Repeat Steps 1-6 for the next set of foils

To ensure the minimisation of biases in the experiments, a few precautions and measures are taken such as:

1. Randomised order of testing within each group of foils
2. Foils are always checked prior to assembly and testing for defects
3. A test run is always done once per experiment day to ensure data collection is as accurate as possible

4. RESULTS

4.1 Overview

In this chapter, the experimental results obtained from data collection to data processing is presented for all the foils. The results and discussion will be split into two, based on the two categories of foils that were used – geometric foils and winglet foils.

For each foil in each shape group and each category, the same variables are studied and investigated. The variables that were extracted are lift (or force), coefficient of lift (C_L), power (P), efficiency (η) and Strouhal number (St). These data were determined through the processed data at flow velocity of $U_\infty = 0.15 \text{ m/s}$.

The foils are identified through a shape-thickness ID where a string indicates the shape, and the number represents the thickness. For example, a geometric circular shape foil of 10mm is identified as “circ 10” and a winglet concave shape foil of 15mm is identified as “conc 15”. The thickness level of 0mm is present in all shapes and corresponds to the data of the reference foil also known as the “flat” foil. The table below shows the naming conventions of all available foils:

Table 1 Naming Convention of Available Foils

Type	Geometric Foils (5-20mm)	Winglet Foils (5-20mm)
Shapes	Rectangular Elliptical Circular Triangular Trapezoid	Rectangular Round Triangular Concave
ID	Rect Elli Circ Tri Trap	Rect Roun Tria Conc

Each variable is represented with bar charts. The x-axis of each chart represents the shapes of the category – each shape having 5 bars each representing the varying thicknesses from 0mm to 20mm . The y-axis represents the corresponding variables (F, C_L, P, η, St).

For both categories, the same discussions will be held, comparing the same variables across the foils within the same category. This will provide a more unbiased results comparison and discussion. However, a cross-comparison will be done to determine if there is any relationship between the two categories.

4.2 Winglet Foil Data

This section will cover the results obtained for the **winglet** shaped foils where the foils feature a flat mid-section, and the name of the shape identifies the shape found at the edges (see Section 3.3.2, Figure 10). The foils in this category are the rectangular,

round, triangular and concave foils with naming conventions *rect*, *roun*, *tria* and *conc*, respectively. The thickness once again ranges from 5mm to 20mm with a reference data of 0mm for each set of foils. A summary of data points can be found in Figure 19 which is an excel sheet of relevant data for the **winglet** shaped foils.

foil ID	C.L	avg force	avg power	avg eff	shape	thickness	avg_T	freq	Sweep	St
flat 0	-0.230713867	-0.000159	0.06955253	0.3017403	flat	0	1.43228	0.698187473	0.304751794	1.4184926
rect 0	-0.230713867	-0.000159	0.06955253	0.3017403	rect	0	1.43228	0.698187473	0.304751794	1.4184926
rect 5	0.243807168	0.000168	0.05690172	0.246832	rect	5	1.68655	0.592926136	0.452833764	1.7899798
rect 10	0.006101375	4.2042E-06	0.04449068	0.1871383	rect	10	1.88216	0.531303635	0.457149957	1.6192362
rect 15	0.220106332	0.00015167	0.03839197	0.1591829	rect	15	2.11878	0.471970492	0.461748527	1.4528779
rect 20	-0.015640384	-1.078E-05	0.03092389	0.1269225	rect	20	2.30751	0.433367817	0.464202689	1.3411367
roun 0	-0.230713867	-0.000159	0.06955253	0.3017403	roun	0	1.43228	0.698187473	0.304751794	1.4184926
roun 5	0.072136004	4.9706E-05	0.08790568	0.3656353	roun	5	1.60497	0.623063272	0.462449113	1.9209004
roun 10	0.267251633	0.00018415	0.08394376	0.3398258	roun	10	1.5588	0.641518175	0.462150051	1.9765177
roun 15	0.354306594	0.00024414	0.06813793	0.2667851	roun	15	1.69423	0.590237295	0.46925648	1.8464845
roun 20	-0.310559779	-0.000214	0.07334927	0.2800727	roun	20	1.67404	0.597355633	0.472783519	1.8827993
tria 0	-0.230713867	-0.000159	0.06955253	0.3017403	tria	0	1.43228	0.698187473	0.304751794	1.4184926
tria 5	0.234635391	0.00016168	0.07838116	0.3350709	tria	5	1.45823	0.685764524	0.456610796	2.0875166
tria 10	0.021008803	1.4476E-05	0.08249454	0.3423548	tria	10	1.4679	0.681247108	0.459087727	2.0850146
tria 15	-0.24403203	-0.0001682	0.07253521	0.2927427	tria	15	1.53757	0.650378598	0.466296879	2.0217967
tria 20	0.243266065	0.00016763	0.0652224	0.2576517	tria	20	1.64554	0.607703009	0.468158265	1.8966746
conc 0	-0.230713867	-0.000159	0.06955253	0.3017403	conc	0	1.43228	0.698187473	0.304751794	1.4184926
conc 5	-0.664347085	-0.0004578	0.08748073	0.3688931	conc	5	1.45031	0.689509929	0.459409566	2.111783
conc 10	-0.102058755	-7.032E-05	0.07566765	0.3126198	conc	10	1.51643	0.659441749	0.462244951	2.0321575
conc 15	0.077196954	5.3194E-05	0.04399685	0.1818192	conc	15	1.86408	0.53645826	0.460570458	1.6471788
conc 20	0.189249046	0.0001304	0.0454453	0.1848025	conc	20	1.87697	0.532773039	0.464392771	1.6494397

Figure 19 Data Points for Winglet Shaped Foils

4.2.1 Strouhal Number Comparison (Winglet)

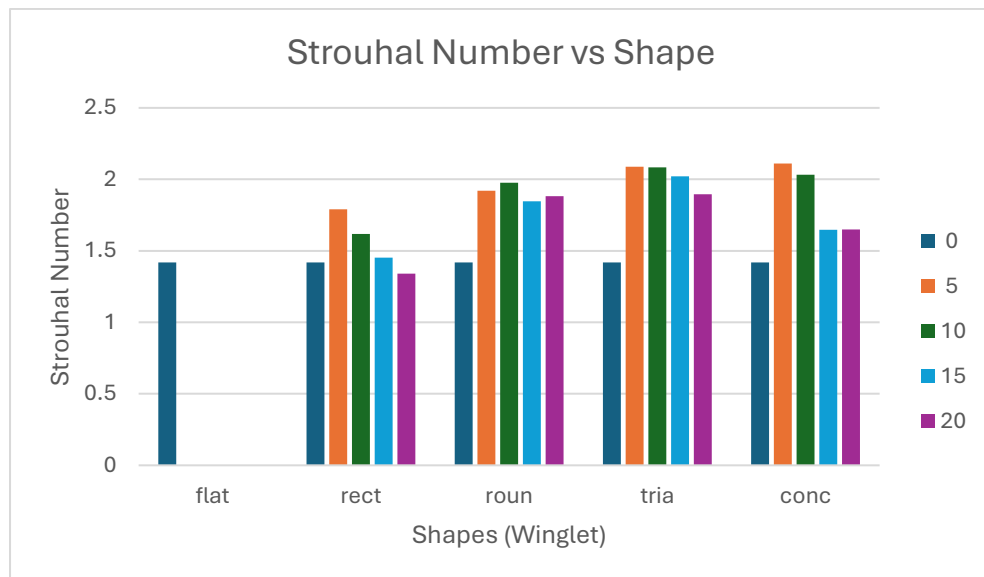


Figure 20 Strouhal Number vs Shape & Thickness (Winglet Foils)

Table 2 Strouhal Number Values (Winglet)

Strouhal Number	Thickness (mm)				
Shape	0	5	10	15	20
Flat	1.418492567	-	-	-	-
Rect	1.418492567	1.789979826	1.619236228	1.452877863	1.341136707
Roun	1.418492567	1.920900381	1.976517713	1.846484502	1.882799321
Tria	1.418492567	2.087516567	2.085014576	2.021796739	1.896674577
Conc	1.418492567	2.111783047	2.032157457	1.647178843	1.649439653

Observing the data points and the bar graph, it shows that the thickness of the foil has a strong effect on the Strouhal number. This reflects the oscillation speed of the foil relative to the incoming flow and a clear relationship can be found: a higher Strouhal number indicates faster oscillation motion which ultimately leads to higher energy extraction. For reference, the baseline Strouhal number found in the flat foil is at $St \approx 1.42$. At optimal thickness, all the winglet shaped foils exceed this value.

Key trends observed by shape:

1. **Rectangular** (*rect*)
 - St drops significantly with increasing thickness
 - Lowest St values among winglets
2. **Rounded** (*roun*)
 - Peaks at $5mm$ with $St \approx 1.9$
 - Remains relatively consistent throughout
 - Indicates fast energetic wake dynamics
3. **Triangular** (*tria*)
 - Relatively high St comparable to concave shape
 - Peaks at $5 - 10mm$ with $St \approx 2.085$
 - Falls off slightly less at $15 - 20mm$ than concave
4. **Concave** (*conc*)
 - Highest St
 - Peaks at $5mm$ where $St \approx 2.1$
 - Remains high at $5 - 10mm$, drops at $15 - 20mm$

As observed, the **rectangular** shape exhibits the lowest Strouhal numbers among the winglet shaped foils, peaking at $St \approx 1.79$ at $5mm$ thickness. This could indicate that increasing the thickness for the rectangular shaped foil potentially dampens the oscillatory responsiveness of the foil. This is consistent with how low-curvature shapes tend to exhibit behaviours that are associated with accumulation of mass and hydrodynamic resistance. With sharp edges and low curvatures, it is expected of the rectangular foil to not only perform worse than other curvier shapes but also reduce in oscillation speed with increased thickness.

On the contrary, the **rounded** foil demonstrates a significantly better performance in unsteady dynamics. It peaks at $St \approx 1.98$ at $10mm$ and remains the most consistent throughout all thicknesses. This suggests that the rounded edge provides smooth formation of wake and remains robust with increasing thickness. For the **triangular** shaped edge foils, they display significantly higher Strouhal numbers, peaking at $St \approx 2.09$ at both $5 - 10mm$. Along with that, they only possess a small, moderate drop in Strouhal number at increased thicknesses of $15 - 20mm$. This could point to sharp / tapered edges being able to support more rapid vortex shedding, thus enhancing oscillatory speed of the foil.

Lastly, the **concave** foil is observed to have produced the largest Strouhal number, peaking at $5mm$ with a value of $St \approx 2.11$. However, with the concave shape, the drop in Strouhal number with increasing thickness is much more drastic than that of the other shapes, dropping down to $St \approx 1.65$ at $15 - 20mm$. This indicates that the concave edge is able to promote higher oscillation speeds. Yet with a rapid decrease, it also points to the fact that this shape is likely to be highly sensitive to

hydrodynamic loading and therefore, may perform best when the thickness is kept to a moderately thin value of 5 – 10mm.

4.2.2 Force Comparison (Winglet)

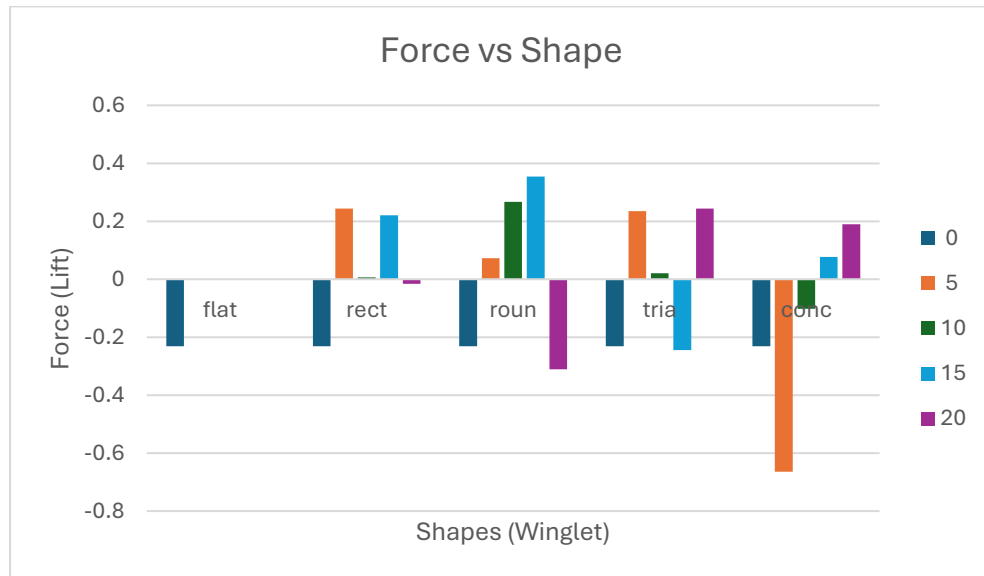


Figure 21 Force vs Shape & Thickness (Winglet Foils)

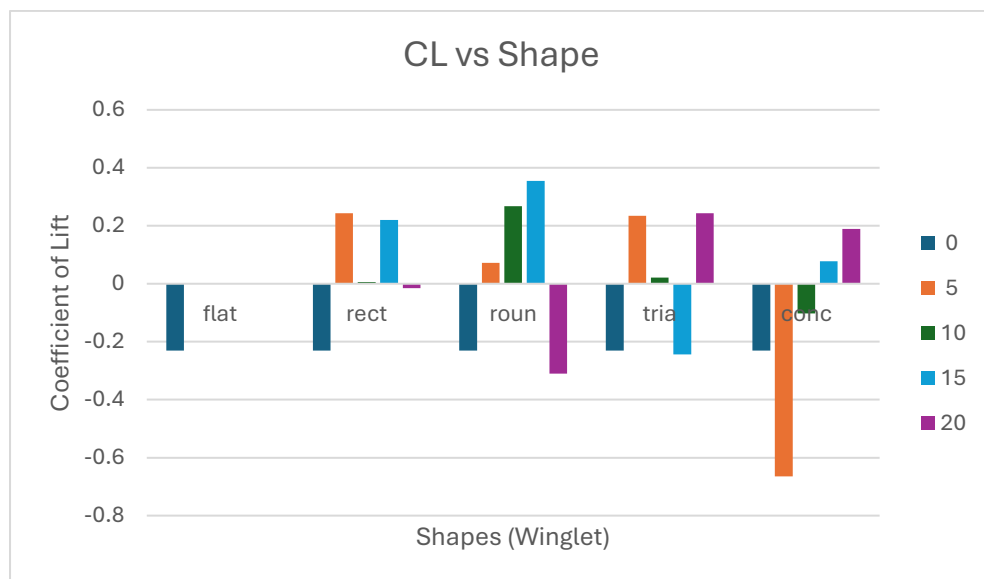


Figure 22 Coefficient of Lift vs Shape & Thickness (Winglet Foils)

As seen in the above figures, the results of force – and therefore coefficient of lift – show substantial variations that does not indicate any clear trend as seen in the Strouhal number observations. The values that are obtained are irregular and fluctuate between positive and negatives across the varying thicknesses and geometry. For example, in the rectangular configuration, the foils generate positive lift at 5mm and 15mm but produce close to 0 *N* or even slightly negative lift at 10mm and 20mm. Even the irregularities do not share a common pattern, leading to unpredictability across all shapes and thickness. Needless to say, these abrupt irregularities do not align with proven relationships between geometry and

oscillation dynamics, therefore contradicting the more consistent patterns found in other metrics of performance.

However, this behaviour and result can be somewhat expected due to the indirect method used to estimate the hydrodynamic lift as opposed to using load cells to get an accurate reading. In this experimental setup, lift is reconstructed using a series of code that utilises foil displacement and damping parameters. Due to the nature of these calculations relying heavily on differentiation of displacement data and in particular, second derivatives, the estimates can be very sensitive to small amount of noise accumulated in the sensors. As observed in other studies as well, this is a common limitation found in flapping foil experiments that are conducted without the use of load cells or force transducers, especially at lower flow velocities where forces present are relatively small, resulting in signal-to-noise ratio becoming more undesirable [7], [11], [24]. In these scenarios, even with data being filtered, minor irregularities in the displacement data collected by the sensors can result in disproportionately large variations in the inferred or reconstructed lift (F) or coefficient of lift (C_L).

As a result, the lift and C_L results that are obtained are considered to be unreliable for the use of comparative analysis. As such, the foil performance will be focused on performance metrics that are more consistent and interpretable such as Strouhal number, power and efficiency. These tend to show more coherent trends across the varying geometry and thickness of the foils and align better with expected behaviour in past research [4], [23], [29]. These will provide a better and more reliable foundation for the evaluation of energy harvesting capabilities and performance.

4.2.3 Power Comparison (Winglet)

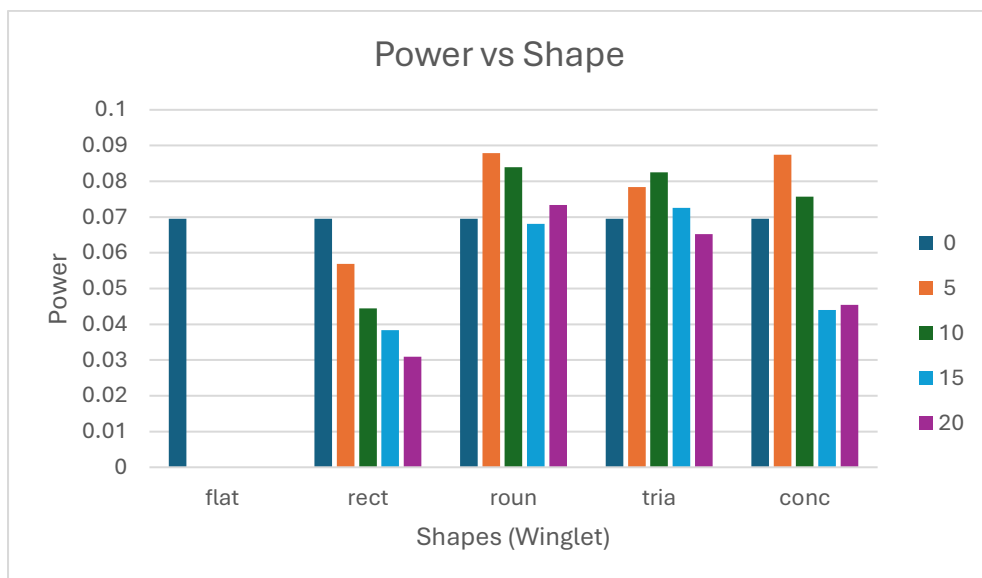


Figure 23 Power vs Shape & Thickness (Winglet Foils)

Table 3 Power Values (Winglet)

Power	Thickness (mm)				
Shape	0	5	10	15	20
Flat	0.069552532	-	-	-	-
Rect	0.069552532	0.056901723	0.044490682	0.038391969	0.030923886
Roun	0.069552532	0.087905675	0.083943755	0.068137928	0.073349272
Tria	0.069552532	0.078381156	0.082494544	0.072535214	0.065222397
Conc	0.069552532	0.08748073	0.075667649	0.04399685	0.045445298

The power data extracted in Watts (W), show distinct trends that are consistent with the Strouhal number behaviours observed in Section 4.2.1. In this small scale, low freestream velocity flapping foil experiment, a reasonable range of $0.03 - 0.09 W$ of power is observed. As seen, the flat foil produces a reference power output of $\sim 0.07 W$.

Consistent with the Strouhal number chart, the **rectangular** foils are observed to produce the least amount of power with a decreasing trend as thickness increases. At $5mm$, the power produced of about $0.057 W$ is a fair bit lower than that of the reference, flat foil. The consistent decline of power output with thickness also suggests that in the case of the rectangular foil, thickening it will primarily introduce inertial and viscous losses. As such, this foil shape can be deemed to be ineffective for energy harvesting applications.

However, it is observed in the **round** foil shape to produce the highest overall power extraction. At $5mm$, the round winglet foil is able to extract an average of $0.088 W$ over the oscillation period. It maintains a similar level of power extraction at $10mm$ and although it falls off slightly at $15mm$ thickness, the power extracted is recovered at $20mm$ at about $0.073 W$. This pattern agrees with its consistent Strouhal numbers and shows that the rounded edge proves to be a reliable, robust foil shape that generates consistent power over a range of thicknesses.

As for the **triangular** winglet foil, it displays a more gradual change in power extraction as thickness varies with a peak of $\sim 0.082 W$ at $10mm$. Beyond this optimal thickness, the power output gradually drops between $15 - 20mm$. Similarly, this trend coincides with the trend observed in the Strouhal number relationship with the triangular foil where it also maintains a relatively high Strouhal number between $5 - 10mm$ before it loses its strength at thicker foils.

Lastly, it is observed that the **concave** foil generates a similar peak power at $5mm$ of approximately $0.087 W$, comparable to the rounded foil. However, it has a steep decline as thickness increases past $5mm$, dropping drastically as thickness approaches $15 - 20mm$. This significant drop in power suggests that the concave profile is highly sensitive to thickness where a thinner foil of this profile can generate considerable power, its performance is greatly impacted when thickness exceeds an optimal range and lead to more destructive effects rather than beneficial ones.

Overall, it can be concluded that the power output generated by the foils can be linked proportionally to the Strouhal number observed for each foil shape and

thickness. A higher Strouhal number tend to also deliver higher power output as observed in the rounded, triangular and concave profiles.

4.2.4 Efficiency Comparison (Winglet)

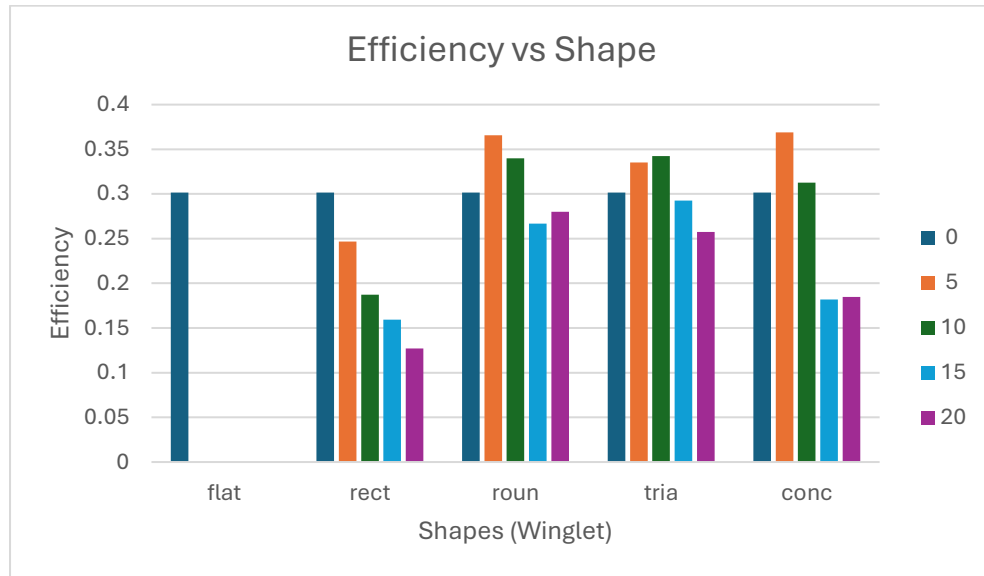


Figure 24 Efficiency vs Shape & Thickness (Winglet Foils)

Table 4 Efficiency Values (Winglet)

Efficiency	Thickness (mm)				
Shape	0	5	10	15	20
Flat	0.301740343	-	-	-	-
Rect	0.301740343	0.246832016	0.187138305	0.159182922	0.126922493
Roun	0.301740343	0.365635304	0.339825832	0.266785074	0.28007266
Tria	0.301740343	0.335070891	0.342354812	0.292742662	0.257651699
Conc	0.301740343	0.368893123	0.312619773	0.18181924	0.184802495

Similar to the power data obtained, the summarised information of the efficiency of each foil shown in Figure 24 also align well with the trends found in the Strouhal number and power trends. A baseline of $\eta = 0.3$ can be found in the flat foil and can be used as a comparison for the other foils. In most cases, a slight increase in thickness results in an increase in efficiency. However, as the foils continue to thicken, the efficiency dips, even dropping to be significantly less efficient than the flat foil.

Repeatedly, it is observed that the **rectangular** foil displays the weakest efficiency results among the winglet foil shapes. Starting at 5mm , it has an efficiency of $\eta = 0.25$ and steadily decreases in efficiency to $\eta = 0.13$ at 20mm . Aligning with Strouhal number and power trends for the rectangular foil, this trend proves not only that increasing the thickness dampens the efficiency but also that the rectangular foil is a highly ineffective foil shape to be used for energy harvesting.

Similar to the power chart, the **rounded** winglet foil exhibits a similar increase-decrease-increase efficiency trend. It peaks at 5mm with $\eta = 0.37$ then drops down

to below the baseline efficiency, though it increases slightly at $20mm$ to come close to the reference efficiency.

The efficiency trend of the **triangular** winglet foil also follows closely to its Strouhal number and power counterparts. Peaking with an efficiency of 0.34 at $10mm$, and the lowest efficiency at $\eta = 0.26$ at $20mm$. This foil shape holds the least difference between maximum and minimum efficiency with a difference of 0.9 . Similar to the round winglet foil, it holds a good efficiency value above the reference foil for an optimal range of $5 - 10mm$ but falls below by a slight amount when the thickness exceeds the ideal range. This also shows that the triangular winglet profile is capable of promoting wake structures that can improve energy transfer from the incoming flow.

Lastly, the **concave** winglet also holds well when thickness complements the foil shape. In fact, the concave winglet profile peaks with the highest efficiency of $\eta = 0.37$ at $5mm$. However, like its Strouhal number and power, as thickness moves past the $5mm$ benchmark, its performance significantly decreases. This once again suggests that though the concave winglet profile is highly effective at generating benefits for energy harvesting, its shape is only effective on the thinner spectrum while its benefits diminish when thickness increases.

Overall, the efficiency graph for the varying shapes and thickness highly complements the Strouhal number and power trends observed for all the tested foils.

4.3 Geometric Foil Data

In this section, the same metrics are discussed for the **geometric** shaped foils as it was in the winglet shaped foils. The geometric shaped foils feature no flat mid-section, and the name of the shape identifies the shape of the foil (see Figure 11). The foils in this category are the **rectangular**, **elliptical**, **circular**, **triangular** and **trapezoidal** foils with naming conventions *rect*, *elli*, *circ*, *tri* and *trap*, respectively. In this set, the rectangular foil that was also used in the winglet shaped data is present as it fits into both categories and presents a method of unbiased testing or benchmarking. Additionally, there are two foil shapes named “triangular”, one belongs to the winglet group while the other in this geometric shaped group. However, in this group, its foil ID is “tri”. (see Figure 10c vs Figure 11d)

The thickness once again ranges from $5mm$ to $20mm$ with a reference data of $0mm$ for each set of foils. A summary of data points can be found in Figure 25 which is an excel sheet of relevant data for the **winglet** shaped foils.

The rectangular foils being a repeat experimental piece, it will not be rediscussed in detail, but its overall trends will be mentioned as comparison to the other foils in this category.

foil ID	C_L	avg force	avg power	avg eff	shape	thickness	avg_T	freq	Sweep	St
flat 0	-0.230713867	-0.000159	0.0695525	0.3017403	flat	0	1.43228	0.698187473	0.304751794	1.41849257
rect 0	-0.230713867	-0.000159	0.0695525	0.3017403	rect	0	1.43228	0.698187473	0.304751794	1.41849257
rect 5	0.243807168	0.000168	0.0569017	0.246832	rect	5	1.68655	0.592926136	0.452833764	1.78997983
rect 10	0.006101375	4.204E-06	0.0444907	0.1871383	rect	10	1.88216	0.531303635	0.457149957	1.61923623
rect 15	0.220106332	0.0001517	0.038392	0.1591829	rect	15	2.11878	0.471970492	0.461748527	1.45287786
rect 20	-0.015640384	-1.078E-05	0.0309239	0.1269225	rect	20	2.30751	0.433367817	0.464202689	1.34113671
elli 0	-0.230713867	-0.000159	0.0695525	0.3017403	elli	0	1.43228	0.698187473	0.304751794	1.41849257
elli 5	0.139020778	9.579E-05	0.0950477	0.3996799	elli	5	1.21417	0.823610754	0.460824506	2.53026679
elli 10	0.340871655	0.0002349	0.0870225	0.351907	elli	10	1.36712	0.731465323	0.4658659	2.27176501
elli 15	0.656935966	0.0004527	0.07571	0.2989036	elli	15	1.57452	0.635114938	0.468885655	1.98530856
elli 20	-0.046659721	-3.215E-05	0.0589641	0.2296824	elli	20	1.69644	0.58946806	0.472135821	1.85539324
circ 0	-0.230713867	-0.000159	0.0695525	0.3017403	circ	0	1.43228	0.698187473	0.304751794	1.41849257
circ 5	0.114087224	7.861E-05	0.0818857	0.3513465	circ	5	1.37234	0.72867995	0.457170678	2.22087405
circ 10	-0.378960662	-0.0002611	0.1126501	0.4527074	circ	10	1.2477	0.801474993	0.466009076	2.48996414
circ 15	-0.308640215	-0.0002127	0.0951408	0.3793815	circ	15	1.3777	0.725849242	0.466604067	2.25789472
circ 20	0.134163714	9.245E-05	0.0860508	0.3301832	circ	20	1.47373	0.678548373	0.473766663	2.14315732
tri 0	-0.230713867	-0.000159	0.0695525	0.3017403	tri	0	1.43228	0.698187473	0.304751794	1.41849257
tri 5	-0.080414587	-5.541E-05	0.0807795	0.3433891	tri	5	1.32016	0.75748229	0.460031168	2.32310309
tri 10	-0.01085384	-7.479E-06	0.0950229	0.3877361	tri	10	1.29465	0.77240824	0.460487757	2.37123025
tri 15	-0.155425314	-0.0001071	0.0892267	0.3583702	tri	15	1.33446	0.749367043	0.466887846	2.3324691
tri 20	0.175791345	0.0001211	0.0608593	0.2460903	tri	20	1.57397	0.635335122	0.463367189	1.96262299
trap 0	-0.230713867	-0.000159	0.0695525	0.3017403	trap	0	1.43228	0.698187473	0.304751794	1.41849257
trap 5	-0.036798646	-2.536E-05	0.0703472	0.2969746	trap	5	1.41498	0.706722212	0.459282922	2.16390295
trap 10	-0.30413186	-0.0002096	0.065968	0.2699013	trap	10	1.55594	0.64269952	0.463128208	1.98434851
trap 15	0.509091098	0.0003508	0.0598629	0.2413178	trap	15	1.62165	0.616657108	0.466700277	1.91862695
trap 20	0.567256236	0.0003909	0.055635	0.2184725	trap	20	1.70522	0.586433962	0.473837168	1.85249472

Figure 25 Data Points for Geometric Shaped Foils

4.3.1 Strouhal Number Comparison (Geometric)

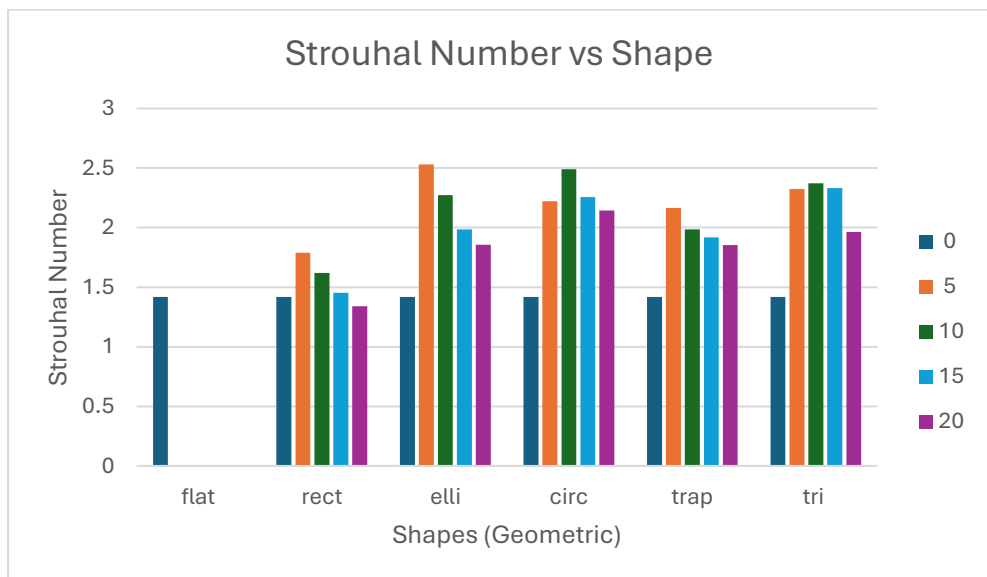


Figure 26 Strouhal Number vs Shape & Thickness (Geometric Foils)

Table 5 Strouhal Number Values (Geometric)

Strouhal Number	Thickness (mm)				
	0	5	10	15	20
Flat	1.418492567	-	-	-	-
Rect	1.418492567	1.789979826	1.619236228	1.452877863	1.341136707
Elli	1.418492567	2.530266794	2.271765008	1.985308558	1.855393241
Circ	1.418492567	2.220874049	2.489964139	2.257894721	2.143157322
Trap	1.418492567	2.163902949	1.984348511	1.918626955	1.852494716
Tri	1.418492567	2.323103086	2.371230255	2.332469096	1.962622995

The bar graph depicts the Strouhal number (St) as a function of thickness for the different shapes of foils. At a glance, it is obvious that these geometric shaped foils tend to produce an overall higher Strouhal number than the winglet shaped foils. This points to the geometric shapes being able to generate higher oscillation speeds and therefore, potentially reaping more benefits in energy harvesting applications.

The Strouhal number trends observed for the foils in the geometric category highlight a distinct yet systematic dependence on foil shape and thickness. These fully shaped foils tend to exhibit much higher oscillatory activity. Using the baseline of $St = 1.42$ found in the flat foil, the geometric foils are observed to show a significant increase in Strouhal number – usually by 40 – 80% -- once thickness is introduced. This highlights the strong influence that a full-foil shape has on the interaction between the foil and flow.

The **rectangular** foil demonstrates the weakest Strouhal number amongst the geometric, fully shaped foils, mirroring its characteristics found in the winglet category. This once again emphasises the weaknesses of the rectangular foil and its relationship between hydrodynamic mass and viscous losses, resulting in slower responses in oscillation especially at higher thicknesses.

The **elliptical** geometric foil is observed to produce the highest Strouhal number, having a peak of $St = 2.53$ at $5mm$. It remains relatively high at lower thicknesses but falls off as thickness approaches $15 - 20mm$, resulting in a Strouhal number of 1.99 and 1.86, respectively. Yet, even at greater thicknesses, the Strouhal number exceeds the baseline by a noticeable amount. This behaviour aligns with the common expectation that smooth and continuously curved geometries found in a foil tend to promote more favourable vortex formation and shedding, which in turn, enables increased oscillatory speeds even at moderate thickness.

The **circular**, or fully rounded, foil also performs well with similar peak values of Strouhal numbers. However, in the case of the circular foil, it peaks at $10mm$ with $St = 2.5$, while it holds a slightly lower value at $5mm$ and $15mm$, though the latter values are quite close with $St = 2.22$ and $St = 2.26$, respectively. The Strouhal number then falls off slightly more at $20mm$ at $St = 2.14$. For this case, the circular foil is able to maintain a Strouhal number of $St > 2$ across all thicknesses. This shows that the circular shaped foil is able to support sustained vortex development as well as strong pressure gradients throughout the cycle of oscillation, therefore outperforming the other geometric shapes.

The **trapezoidal** foil holds a resemblance to the triangular (*tria*) foil in the winglet category; however, it has a smaller flat mid-section with smoother edges. As such, even though the trend observed between the two are similar, the values tend to peak and dip differently. The trapezoidal foil peaks at a relatively high value of $St = 2.16$ at $5mm$, but shows a gradual decline in the Strouhal number as thickness increases. Though it peaks at lower values compared to the other shapes, it demonstrates a relatively stable and consistent behaviour across all thicknesses. This suggests that the trapezoidal geometry is able to maintain oscillatory strength and that its relatively less smooth geometry still provides a balance between sharp-edge acceleration and smooth-edge stability.

The **triangular** (tri) foil here refers to a fully triangular shaped foil with the apex at the middle of the chord (see Figure 11d). This shape demonstrates high Strouhal numbers ranging from $2.32 < St (tri) < 2.37$ at $5 - 15mm$. Even though the Strouhal number drops noticeably to 1.96 at $20mm$, it is able to maintain a very consistent Strouhal number across most thickness, indicating the ability to maintain some of the fastest oscillatory motions across the dataset. This is consistent with how it has been reported that sharp LE can lead to the enhancement of LEV suction and thus, contributes to stronger shedding of vortex, particularly at moderate thicknesses.

4.3.2 Force Comparison (Geometric)

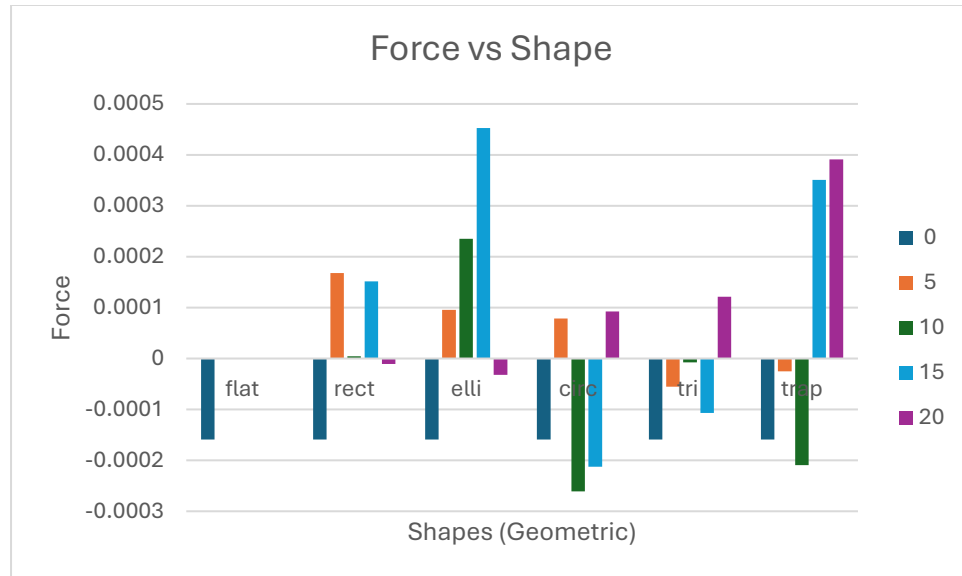


Figure 27 Force vs Shape & Thickness (Geometric Foils)

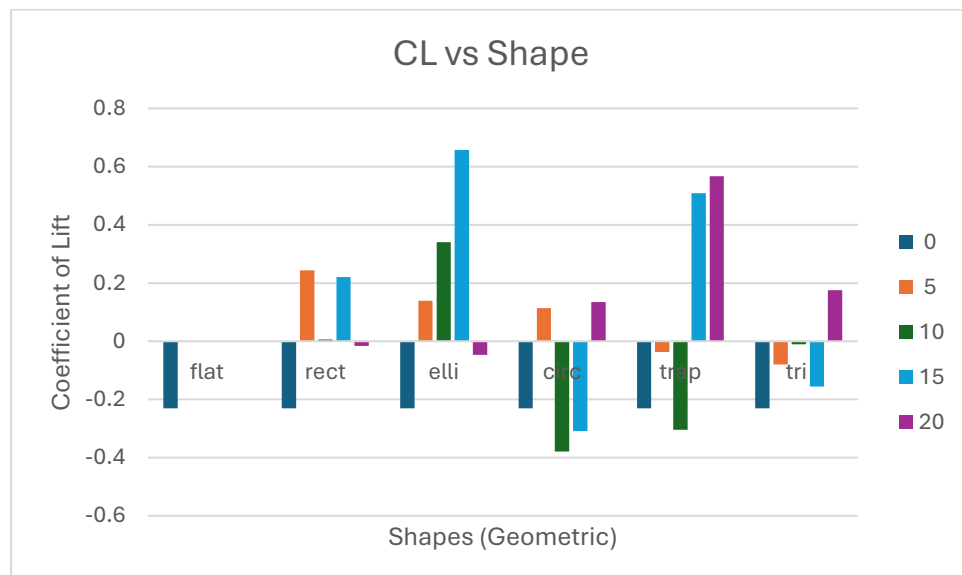


Figure 28 Coefficient of Lift vs Shape & Thickness (Geometric Foils)

As previously observed in the winglet category, a similar observation in force and coefficient of lift (C_L) is observed here in the geometric category as well. The lift and its corresponding coefficient exhibit significant irregularities and fluctuations

across all geometry and thickness. As seen in Figures 27 and 28, the inconsistencies such as sign changes found are unpredictable and frequent, while the magnitudes vary significantly even within a group of foil shapes that are expected to behave in similar fashion across the different thicknesses such as in the elliptical, circular and trapezoidal foils.

As expressed earlier in Section 4.1.2 of the winglet category, this behaviour can be contributed to the use of indirect methods for the estimation of force and subsequently, coefficient of lift. Without load cells or force transducers, the force obtained is reconstructed using the displacement data obtained through the sensors as well as dynamic models that are damping-based. This method is highly sensitive to noise present in the position signals particularly in the data processing segment due to differentiation methods used and second derivatives which amplify these seemingly small noises and irregularities.

However, in the case of geometric, fully shaped foils, they present additional sources of variations that may not be as prominent in the winglet category. The geometric foils tend to present a larger wetted surface area and stronger 3D flow structures which could contribute to increased spatial variations. While these variations and data captured are genuine, the methods used to capture these data such as the indirect displacement-based method cannot capture them as accurately as load cells or force transducers can. In addition to that, the vortex shedding found in shaped foils tend to be stronger and energetic wake interactions are observed more than in the winglet category. This can further affect the sensitivity to minor fluctuations when inferring the force.

As such, using the force and coefficient of lift results for comparison across the foils and categories do not present to be reliable and ideal.

4.3.3 Power Comparison (Geometric)

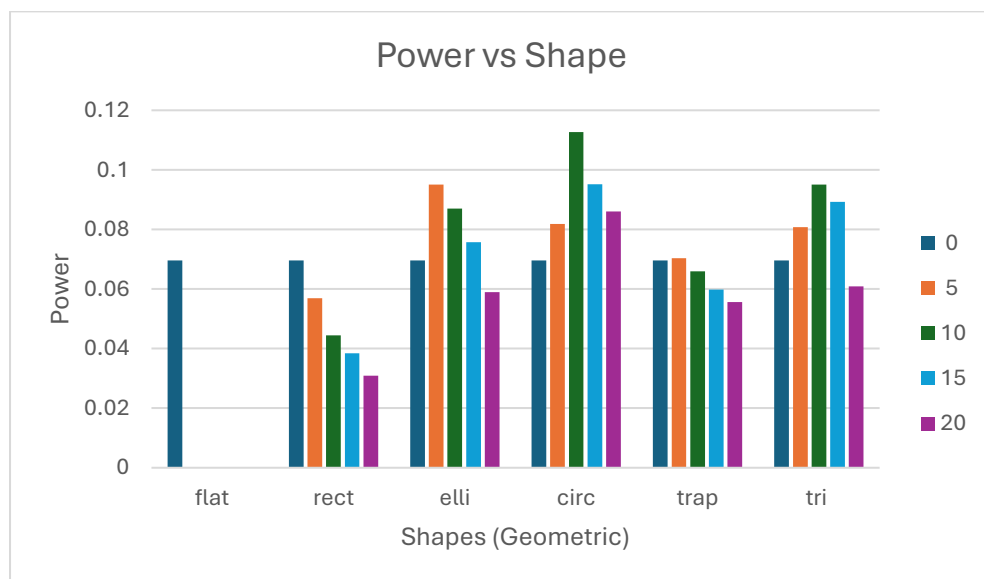


Figure 29 Power vs Shape & Thickness (Geometric Foils)

Table 6 Power Values (Geometric)

Power	Thickness (mm)				
Shape	0	5	10	15	20
Flat	0.069552532	-	-	-	-
Rect	0.069552532	0.056901723	0.044490682	0.038391969	0.030923886
Elli	0.069552532	0.095047654	0.087022545	0.075710011	0.058964122
Circ	0.069552532	0.081885669	0.112650066	0.095140829	0.086050771
Trap	0.069552532	0.070347242	0.065968024	0.05986286	0.055635005
Tri	0.069552532	0.080779451	0.095022922	0.089226734	0.060859297

The power results found in the geometric shaped foils reveal interpretable trends throughout the varying shapes and thicknesses of the foils. As with the winglet category, the flat foil provides a baseline to be compared against with a power extraction of $P = 0.07 W$. Generally, the geometric shaped foils are observed to extract more power with more coherent results as compared to the winglet foils.

The **rectangular** foil as previously noted, reveals to have the weakest power extraction amongst the foil shapes in this category across all thicknesses. It is seen to produce less power compared to the baseline even at small thicknesses, and it continues to decrease as thickness increases. This mirrors its Strouhal number trend and performance, suggesting that the rectangular geometry continues to degrade in unsteady flow, and the performance only weakens with increased thickness. This emphasises the rectangular foil and its incapability to produce favourable and coherent hydrodynamic behaviours for energy harvesting applications.

On the contrary, the **elliptical** foils are observed to demonstrate strong power extraction, with a peak of $0.095 W$ at $5mm$. The power trend observed for the elliptical foil gradually decreases as thickness increases but remains mostly above the baseline power extraction. It only drops below the baseline at $20mm$, producing a power of $0.059 W$. The stability and magnitude of power observed here correspond well to the trends observed in the Strouhal number chart.

The **circular** foil is also observed to perform well above average compared to the baseline output and other foil shapes. At $5mm$, the improvement in power output compared to the baseline foil is minimal, but as it approaches its more optimal thickness at $10mm$, a large spike in power output is observed at $P = 1.13 W$. As thickness increases, it is observed that it moves away from its ideal thickness and hydrodynamic mass, leading to a gradual loss in power extraction at $15 - 20mm$. This performance trend highlights the robustness of the circular foil and also how the circular curvature of the foil is able to support the wake dynamics present.

The trapezoidal foil is observed to produce more mediocre results. However, despite its relatively lower output and gradual decrease below the baseline, the trapezoidal foil is arguably the most consistent in power output amongst the geometric foils, going from $0.07 W$ at $5mm$ to $0.056 W$ at $20mm$. This shows that despite its Strouhal number trends, the trapezoidal geometry of the foil may not provide the

most ideal shape in extracting energy, only benefiting partially from its shape and aerodynamic characteristics at relatively thin foils.

Lastly, the triangular foil has a performance that gradually improves with thickness up to 10mm before falling off as thickness moves beyond an ideal point. It peaks at 10mm and remains largely the consistent at 15mm with powers of 0.095 W and 0.089 W , respectively. The power output decreases more noticeably when it reaches 20mm to produce a slightly below average power of 0.061 W . This is likely due to the triangular shape having a relatively sharper edge that facilitates stronger formations of vortex at moderate thicknesses $10 - 15\text{mm}$). However, as thickness increases further to 20mm and beyond, the geometry and its features may contribute more to an increase of drag or flow separation occurring earlier.

4.3.4 Efficiency Comparison (Geometric)

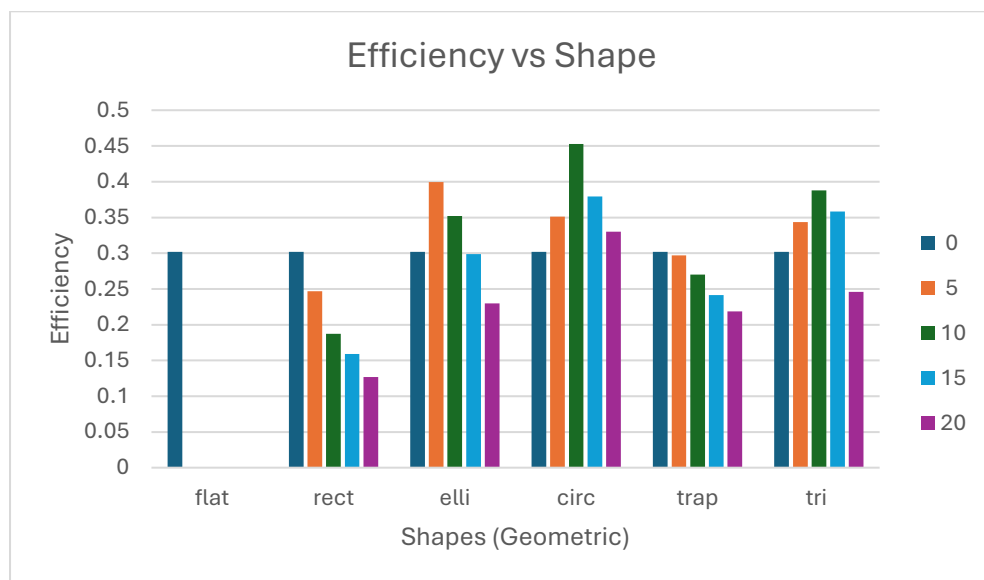


Figure 30 Efficiency vs Shape & Thickness (Geometric Foils)

Table 7 Efficiency Values (Geometric)

Efficiency	Thickness (mm)				
Shape	0	5	10	15	20
Flat	0.301740343	-	-	-	-
Rect	0.301740343	0.246832016	0.187138305	0.159182922	0.126922493
Elli	0.301740343	0.399679876	0.351906985	0.298903561	0.229682398
Circ	0.301740343	0.351346466	0.452707399	0.379381494	0.330183234
Trap	0.301740343	0.296974558	0.269901315	0.241317828	0.218472468
Tri	0.301740343	0.343389102	0.387736146	0.358370157	0.246090337

In the efficiency comparison, the trends and results mimic the observed trends in Strouhal number and power comparisons, providing clear and distinct results for the various shapes and thickness. With a benchmark of $\eta = 0.3$ observed in the flat foil, most foil shapes with most thicknesses are observed to produce a better efficiency, particularly at the moderate to lower thicknesses of $5 - 10\text{mm}$.

Like all the other metrics, the efficiency of the **rectangular** foils are proof that this geometry is the worst performing shape across all thicknesses. With an efficiency of 0.25 at 5mm, and a steady decrease in efficiency at $\eta = 0.13$ at 20mm, well below the baseline. This reiterates that the rectangular geometry is fundamentally unsuitable for applications in energy harvesting.

The **elliptical** shape on the other hand, shows a significantly higher efficiency especially at moderate thickness, hitting a peak efficiency of $\eta = 0.4$. The trend is observed to gradually decrease just like its Strouhal number and power performance but is maintained close to or above the baseline efficiency until the thickness exceeds the operating range giving an efficiency well below the baseline of $\eta = 0.23$ at 20mm.

The circular foil once again proves to be one of the more suitable foil shapes for energy harvesting applications. It demonstrates a steady increase in efficiency in the moderate ranges of thickness and peaking at 10mm with $\eta = 0.45$. This efficiency is considerably high given the Betz Limit that limits the maximum efficiency to $\eta = 0.593$. Despite a decrease in efficiency at the larger two thicknesses, the circular foils are able to maintain an efficiency above the baseline even at high thickness levels. The high efficiency recorded corresponds well to the high Strouhal numbers observed for the circular foil, indicating its ability to sustain high oscillation speeds, making it a suitable candidate for harvesting energy efficiently.

Observing the trapezoidal foils, they tend to produce a below average efficiency and power despite its above average Strouhal numbers. This shows how the trapezoidal despite performing better than the rectangular foil, is unable to match the performance and aerodynamic advantages found in the other foil shapes such as elliptical, circular and even triangular. The mixed shape geometry of a trapezoid is able to provide some benefits but does not counter the lack of consistency that the shape contributes to due to its inconsistent and less smooth curvature that leads to its inability to sustain high efficiency and power wake dynamics.

Lastly, the triangular foil is observed to perform similar to the elliptical foil but peaking at a slightly lower efficiency of 0.39 and at a higher thickness of 10mm. It demonstrates a relatively strong and consistent performance, and it could stem from the benefits of the tapered edges that potentially enhance suction and promote the generation of stronger vortex. Once again, the efficiency drastically drops at higher thickness down to $\eta = 0.25$ at 20mm, signalling that beyond an ideal range, there is too much excess mass and possible premature flow separation due to the large thickness.

4.4 Summary

Across the experiments conducted at $U_{\infty} = 0.15 \text{ m/s}$ in the water tunnel, the performance metrics measured illustrated clearly that the flapping foils showed sensitivity to foil geometry as well as thickness.

It was observed firstly that the geometric or fully shaped foils tend to outperform the winglet shaped foils across all performance metrics. Secondly, it was also observed that moderate thickness was a key to unlocking optimal performances for every foil shape.

In both categories, the rectangular foil was consistently underperforming compared to the other foils, showing low values of Strouhal number, Power and Efficiency.

In the winglet category, the round, triangular and concave edged foils produced relatively higher Strouhal numbers, stronger power, and higher efficiencies within moderate ranges of thickness (5 – 10mm), where they peaked in these metrics. These show how modifying and shaping the edges of the foils in itself can enhance wake dynamics. However, as observed, the performance enhancements are limited by the large area of the flat mid-section, constraining the overall vortex development and possibly causing premature flow separation.

The geometric category proved to have a much better performance by a solid margin. The elliptical, triangular and particularly the circular foil shapes achieved exceptionally high values for the performance metrics discussed. They maintained strong performances across multiple thicknesses and demonstrated that a foil with a fully shape geometry is more likely to be able to sustain coherent vortex shedding and more energetic wake interactions between the foil and flow. This could be attributed to the smoother geometry with less abrupt edges allowing flow to also separate more naturally.

The force and coefficient of lift data proved to be inconsistent and noisy with its cause attributed to indirect reconstruction of force as well as the low flow velocities. These can lead to high sensitivity to noise despite filtering and as such, are deemed to be unreliable for the comparison of metrics among the foils.

5. CONCLUSION

5.1 Summary of Work

To reiterate, this study investigated how geometry and thickness influenced a foil with the aim of finding the optimal geometry and thickness for energy harvesting applications. Conducted in a low-speed water tunnel, two main categories of foils were experimented on and examined: (1) Winglet shaped foils, (2) Geometric or fully shaped foils. A total of 8 pairs of shaped foils and 1 pair of flat, reference foils were 3D-printed with PLA, varying in shape and thickness.

Oscillatory motion and its displacement was measured using sensors and were processed using MATLAB to retrieve the key performance metrics discussed in Section 4: Strouhal number, Power and Efficiency. Due to limitations of indirect force reconstruction methods, the metrics of force and coefficient of lift was omitted for the use of comparison between the foils.

This study compared how the different geometrical features (e.g. flat mid-sections in the winglet category, or smoother curvatures in the geometric category) influenced wake dynamics and the foil's capability in energy harvesting.

5.2 Key Findings

5.2.1 Geometric Influence

Based on the performance metrics discussed in Section 4 and the comparison done between the two foil geometry categories, it was found that the foils in the geometric category – those fully shaped – outperformed those in the winglet category substantially across most, if not all the metrics. It was found that the geometric shaped foils tend to produce significantly higher Strouhal numbers, power output as well as superior efficiencies compared to their counterparts in the winglet category.

As previously mentioned, the rectangular shaped foil, whose data points were used in both categories, demonstrated comparatively poor performance across all metrics. This indicates that highly angular and right-angled profiles are likely to inhibit beneficial characteristics in unsteady flow.

5.2.2 Thickness Influence

A common observation across both categories and all shapes is that the performance metrics measured usually peak at low to moderate thickness (5 – 10mm). This indicates that at these levels of thickness, several factors are enhanced:

- Strouhal numbers are peaking
- Increased oscillation speeds / frequencies
- Peak power extractions
- Peak efficiency levels of up to 45% in the circular foils

The other common trend was that past this optimal range of moderate thickness (15 – 20mm), performance was greatly hindered and declined due to a variety of reasons that are obvious such as added mass, as well as the unseen factors like hydrodynamic damping or less coherent vortex.

5.2.3 Limitations for Measurement

Since there was no equipment such as force transducers and load cells, this experiment set up was unable to retrieve force output directly, and consequently, the coefficient of lift was unable to be obtained directly too. As a result, an indirect reconstruction of these metrics was used to obtain the values. However, given the low velocity and hypersensitivity of the setup to noise, the values obtained were irregular with no specific trend observed. It is worth to note that even so, this did not affect the other performance metrics obtained.

5.3 Implications for Energy Harvesting Applications

The findings in this study provides potential insights into the design of flapping foils – bio-inspired or mechanically actuated – for energy harvesting:

1. Fully shaped / Geometric foils are more likely to influence results positively compared to edge-only shape modifications

2. Curved and tapered planforms (e.g. circular, elliptical, triangular) tend to promote more stable vortex formations and consequently lead to increased frequencies in oscillations
3. Balancing between structural integrity, mass and performance lies in finding the ideal thickness of the foil
4. Blocky and highly angular geometries are extremely inefficient and unsuitable for energy harvesting

These observations are reiterations of current observations in existing literature on bio-inspired propulsion and vortex-induced energy mechanisms [4], [23], [29].

5.4 Limitations of This Study

Despite a fair amount of meaningful and insightful results obtained that are mostly consistent, there are some limitations that are present in this study:

- Indirect reconstruction of force and thereafter, coefficient of lift limited the reliability and usability of the lift and C_L data points
- Low freestream velocity leads to increased proneness to noise in the displacement data measured by the sensors
- Using only one material – PLA – allowed for consistency but limited the experiment in the ability to test the possible effects of material
- Visualisation of the wake through methods such as PIV was not conducted, hence direct observation was not achieved of the vortex dynamics

5.5 Recommendations for Future Work

Building upon this study and the limitations identified, the following are some recommendations for future studies:

1. **Use of direct force measurements:** the use of loads cells, strain-gauge sensors or other types of force transducers can aid in obtaining reliable and usable data on lift (L), thrust (T) and moment (M).
2. **Use of PIV or dye injection:** this allows the visualisation of the wake structures and therefore the direct linking of it with performance.
3. **Use of flexible / bio-inspired materials:** allows the possible broadening of effective operating envelope since flexibility may lead to the improvement of LEV attachment.
4. **Use of higher freestream velocities:** this can assist in reducing noise and thus improving signal-to-noise ratio.
5. **Use of Computational Fluid Dynamics (CFD) simulations:** CFD simulations can aid in validating test results and observations, allowing more detailed analysis to be conducted.

REFERENCES

- [1] K. K. Jaiswal, C. R. Chowdhury, D. Yadav, R. Verman, S. Dutta, K. S. Jaiswal, B. Sangmesh and K. S. K. Karuppasamy, "Renewable and Sustainable Clean Energy Development and Impact on social, economic, and Environmental Health," *Energy Nexus*, vol. 7, no. 7, pp. 100-118, September 2022.
- [2] International Energy Agency, "Net Zero by 2050: A Roadmap for the Global Energy Sector," IEA, Paris, 2021.
- [3] Ocean Energy Systems, "Ocean Energy and Net Zero: An International Roadmap to Develop 300GW of Ocean Energy by 2050," OES, 2023.
- [4] M. Triantafyllou, G. Triantafyllou and D. K. P. Yue, "Hydrodynamics of Fishlike Swimming," *Annual Review of Fluid Mechanics*, vol. 32, no. 1, pp. 33-53, January 2000.
- [5] Q. Zhu, "Optimal Frequency for Flow Energy Harvesting of a Flapping Foil," *Journal of Fluid Mechanics*, vol. 675, pp. 495-517, 25 May 2011.
- [6] J. M. Anderson, K. Streitlien, D. S. Barrett and M. S. Triantafyllou, "Oscillating Foils of High Propulsive Efficiency," *Journal of Fluid Mechanics*, vol. 360, pp. 41-72, 1998.
- [7] L. E. Muscutt, G. D. Weymouth and B. Ganapathisubramani, "Performance augmentation mechanism of in-line tandem flapping foils," *Journal of Fluid Mechanics*, vol. 827, pp. 484-505, 2017.
- [8] M. Sfakiotakis, D. Lane and J. B. C. Davies, "Review of fish swimming modes for aquatic locomotion," *IEEE Journal of Oceanic Engineering*, vol. 24, no. 2, pp. 237-252, April 1999.
- [9] N. S. Lagopoulos, B. Ganapathisubramani and G. D. Weymouth, "Deflected wake interaction of tandem flapping foils," *Journal of Fluid Mechanics*, vol. 903, no. 1, p. A9, 2020.
- [10] I. Akhtar, R. Mittal, G. V. Lauder and E. Drucker, "Hydrodynamics of a biologically inspired tandem flapping foil configuration," *Theoretical and computational fluid dynamics*, vol. 21, no. 3, pp. 155-170, 2007.
- [11] K. Sampath, J. D. Geder, R. Ramamurti, M. D. Pruessner and R. Koehler, "Hydrodynamics of tandem flapping pectoral fins with varying stroke phase offsets," *Phys. Rev. Fluids*, vol. 5, no. 9, pp. 94-101, September 2020.
- [12] J. Alaminos-Quesada and R. Fernandez-Feria, "Aerodynamics of Heaving and Pitching Foils in Tandem from Linear Potential Theory," *AIAA journal*, vol. 58, no. 1, pp. 37-52, 2020.
- [13] N. Kumari, A. Chakraborty and S. Jangam, "The hydrodynamic analysis of multiple hydrofoils translating in tandem in presence of a free surface," *Proceedings of the Institution of Mechanical Engineers. Part M, Journal of engineering for the maritime environment*, vol. 237, no. 4, pp. 1008-1026, 22 June 2022.
- [14] J. Li, P. Wang, X. An, D. Lyu, R. He and B. Zhang, "Investigation on Hydrodynamic Performance of Flapping Foil Interacting with Oncoming Von Kármán Wake of a D-Section Cylinder," *Journal of Marine Science and Engineering*, vol. 9, no. 6, p. 658, June 2021.
- [15] E. E. Handy-Cardenas, Y. Zhu and K. S. Breuer, "Optimal kinematics for energy harvesting using favourable wake-foil interactions in tandem oscillating hydrofoils," *Journal of Fluid Mechanics*, vol. 1012, p. A23, 9 June 2025.
- [16] K. D. Jones, C. M. Dohring and M. F. Platzer, "Experimental and Computational Investigation of the Knoller-Betz Effect," *AIAA journal*, vol. 36, no. 7, p. 1240-1246, July 1998.
- [17] D. Floryan, T. V. Buren, C. W. Rowley and A. J. Smits, "Scaling the propulsive performance of heaving and pitching foils," *Journal of Fluid Mechanics*, vol. 822, pp. 386-397, June 2017.
- [18] T. V. Buren, D. Floryan and A. J. Smits, "Scaling and Performance of Simultaneously Heaving and Pitching Foils," *AIAA Journal*, vol. 57, no. 9, p. 3666-3677, March 2018.
- [19] J. A. Bowker, N. C. Townsend, M. Tan and R. A. Shenoi, "Experimental study of a wave energy scavenging system onboard autonomous surface vessels (ASVs)," *OCEANS 2015 - Genova*, pp. 1-9, 2015.
- [20] F. F. Siala and J. A. Liburdy, "Power estimation of flapping foil energy harvesters using vortex impulse theory," *Renewable Energy*, vol. 154, pp. 894-902, 2020.
- [21] Y.-J. Liou, B. Lan, C.-Y. Chiang and Y.-H. Lai, "Strouhal number impact on propulsion efficiency in fully-active oscillating foils: Unpacking frequency, amplitude, and velocity relationships," *Ocean engineering*, vol. 293, pp. 116-686, February 2024.

- [22] B. Sun, W. Li, Z. Wang, Y. Zhu, Q. He, X. Guan, G. Dai, D. Yuan, A. Li, W. Cui and D. Fan, "Recent Progress in Modeling and Control of Bio-Inspired Fish Robots," *Journal of Marine Science and Engineering*, vol. 10, no. 6, p. 773, June 2022.
- [23] G. K. Taylor, R. L. Nudds and A. L. R. Thomas, "Flying and swimming animals cruise at a Strouhal number tuned for high power efficiency," *Nature*, vol. 425, no. 6959, pp. 707-711, 16 October 2003.
- [24] T. Kinsey and G. Dumas, "Parametric Study of an Oscillating Airfoil in a Power-Extraction Regime," *AIAA*, vol. 46, no. 6, pp. 1318-1330, June 2008.
- [25] L. Schouveiler, F. S. Hover and M. S. Triantafyllou, "Performance of flapping foil propulsion," *Journal of Fluids and Structures*, vol. 20, no. 7, pp. 949-959, October 2005.
- [26] P. K. Yadav, A. Kumar and S. Jaiswal, "A critical review of technologies for harnessing the power from flowing water using a hydrokinetic turbine to fulfill the energy need," *Energy Reports*, vol. 9, pp. 2102-2117, December 2023.
- [27] V. Joshi and R. C. Moya, "Mechanism of wake-induced flow dynamics in tandem flapping foils: Effect of the chord and gap ratios on propulsion," *Physics of fluids*, vol. 33, no. 8, pp. 87-104, August 2021.
- [28] D. N. Beal, F. S. Hover, M. S. Triantafyllou, J. C. Liao and G. V. Lauder, "Passive propulsion in vortex wakes," *Journal of Fluid Mechanics*, vol. 549, pp. 385-402, 8 February 2006.
- [29] Z. Peng and Q. Zhu, "Energy harvesting through flow-induced oscillations of a foil," *Physics of Fluids 21*, vol. 21, no. 12, pp. 123602-123602-9, 22 December 2009.
- [30] K. Isogai, Y. Shinmoto and Y. Watanabe, "Effects of Dynamic Stall on Propulsive Efficiency and Thrust of Flapping Airfoil," *AIAA*, vol. 37, no. 10, pp. 1145-1151, October 1999.
- [31] M. Ashraf, J. Young and J. Lai, "Reynolds number, thickness and camber effects on flapping airfoil propulsion," *Journal of Fluids and Structures*, vol. 27, no. 2, pp. 145-160, February 2011.
- [32] SKYbrary, "SKYbrary," 29 December 2020. [Online]. Available: <https://skybrary.aero/articles/camber>. [Accessed 17 October 2025].
- [33] R. Godoy-Diana, J.-L. Aider and J. E. Wesfreid, "Transitions in the wake of a flapping foil," *Phys. Rev. E*, vol. 77, no. 1, p. 016308, January 2008.
- [34] S. Fu and S. Raayai-Ardakani, "Double-light-sheet, consecutive-overlapping particle image velocimetry for the study of boundary layers past opaque objects," *Experiments in fluids*, vol. 64, no. 11, October 2023.
- [35] I. M. Viola, G. Pisetta, W. Dai, A. Arredondo-Galeana, A. Young and A. Smyth, "Morphing Blades: Theory and Proof of Principles," *International marine energy journal*, vol. 5, no. 2, pp. 183-193, October 2022.
- [36] P. K. Swain, S. P. Dora and A. K. Barik, "Energy extraction performance of tandem flapping foil undergoing elliptical motion trajectory," *Ocean engineering*, vol. 268, pp. 113-390, January 2023.
- [37] G. Wang, C. Zhao and B. F. Ng, "Interlinked tandem hydrofoils for synergistic energy harvesting and triboelectric power generation," *Energy Conversion and Management*, vol. 348, no. B, pp. 120-712, 15 January 2026.

APPENDIX A: Foil Details

This section provides a detailed breakdown of all tested foils. Each foil has a chord length (c) of $0.1m$ and a span (b) of $0.5m$. Figure 31 shows the rectangular foil of which the data was used in both categories. Figures 32, 33, 34 show the foils found in the winglet category – round, triangular and concave. Figures xx show the foils found in the geometric category – elliptical, circular, triangular, trapezoidal.

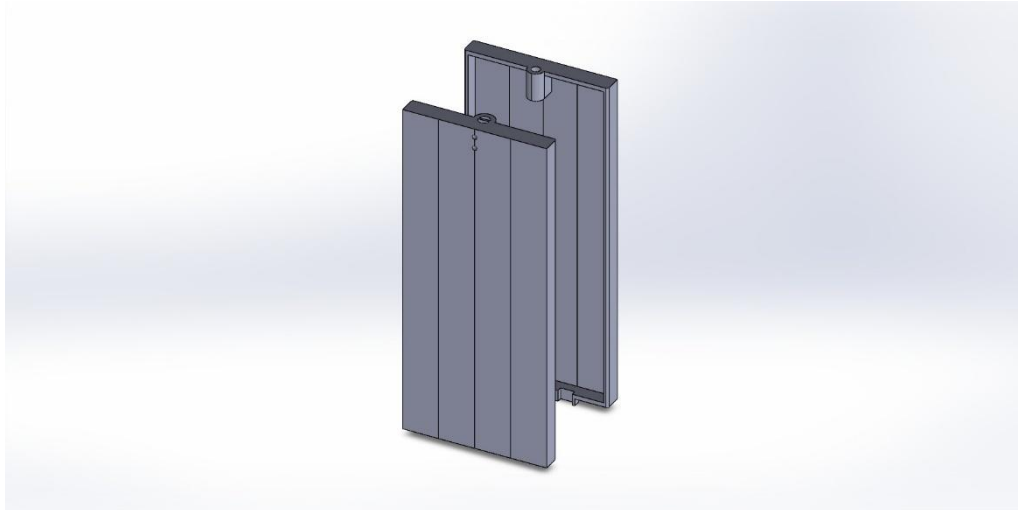


Figure 31 Rectangular Foils – “Rect” (Isometric View)

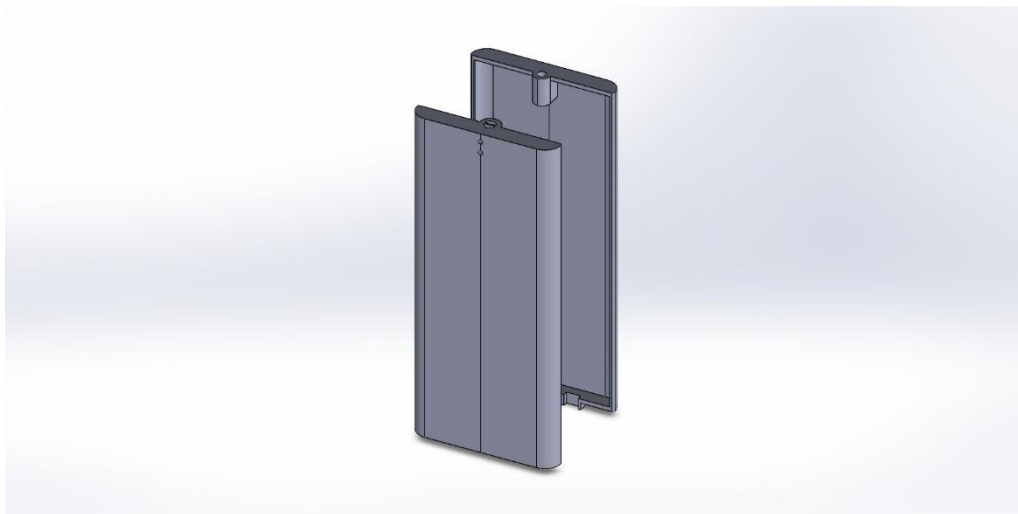


Figure 32 Round Foils – “Roun” (Isometric View)

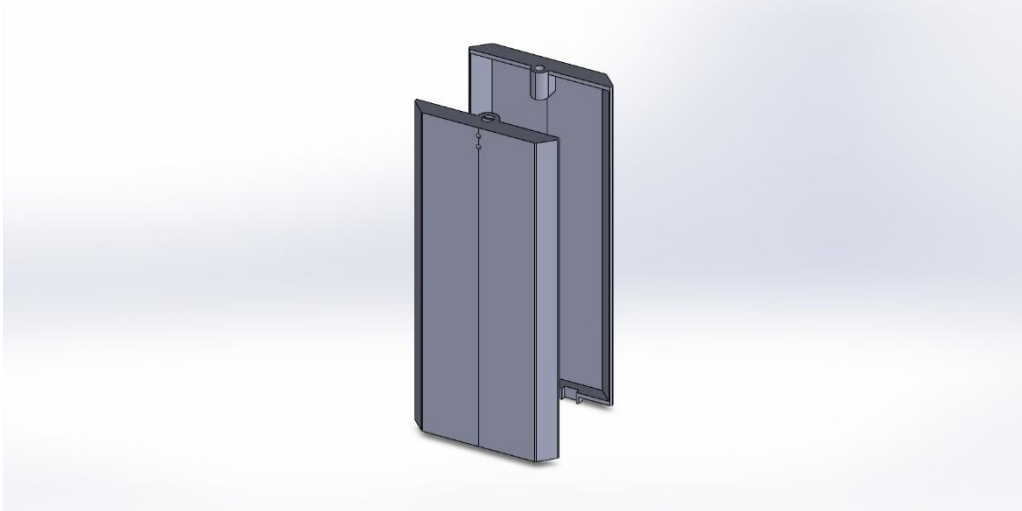


Figure 33 Triangular Foils – "Tria" (Isometric View)

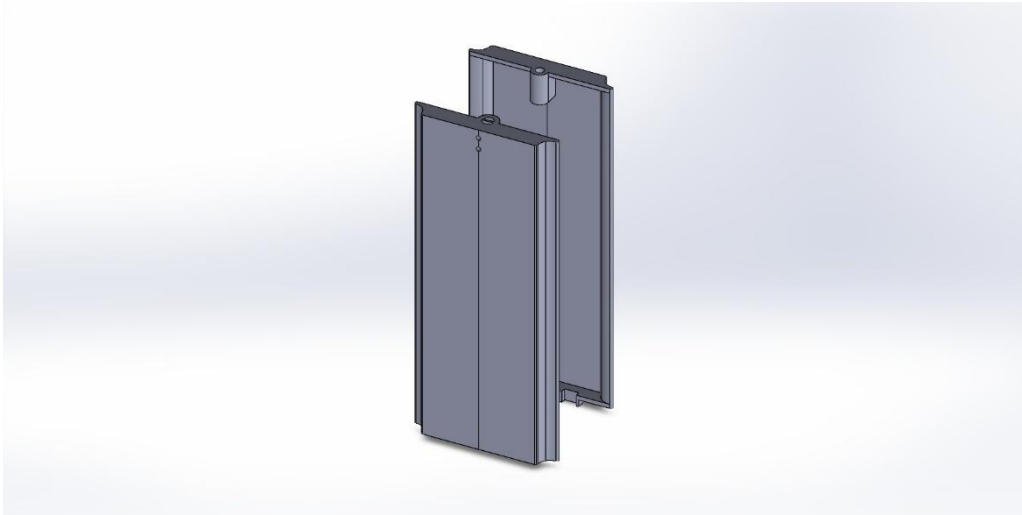


Figure 34 Concave Foils – "Conc" (Isometric View)

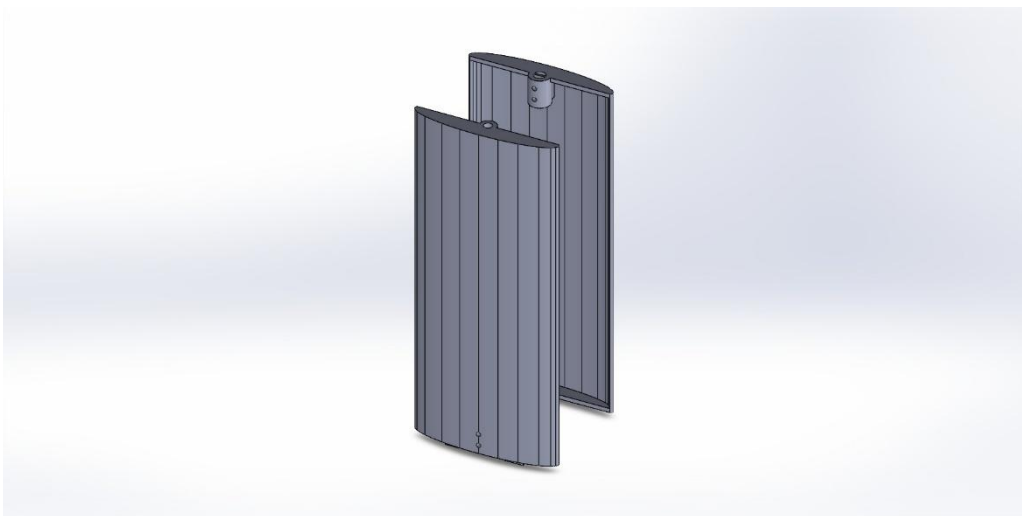


Figure 35 Elliptical Foils – "Elli" (Isometric View)

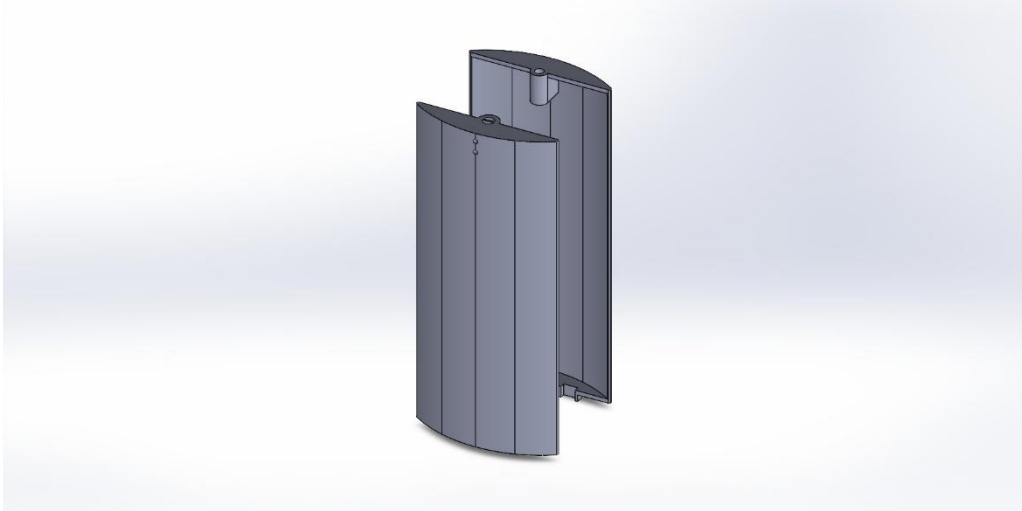


Figure 36 Circular Foils – “Circ” (Isometric View)

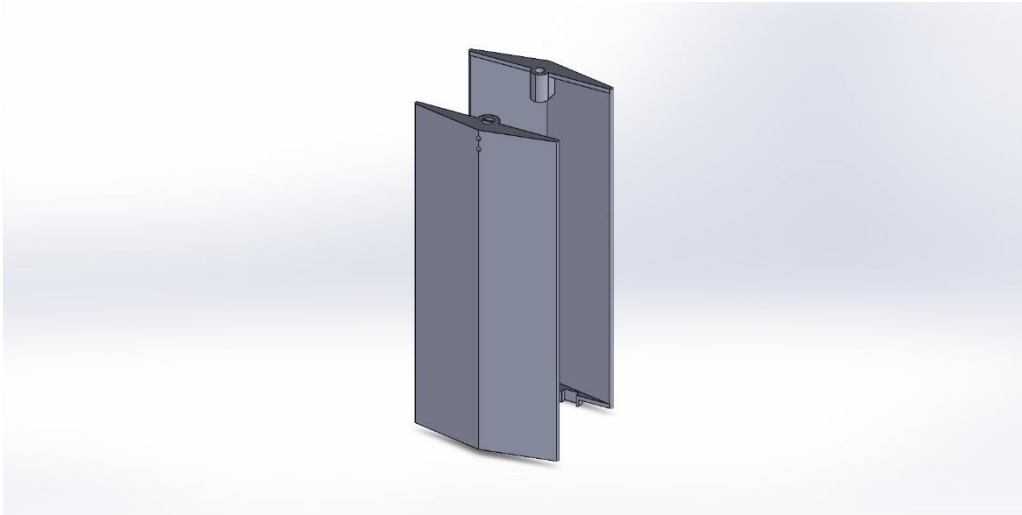


Figure 37 Triangular Foils – “Tri” (Isometric View)

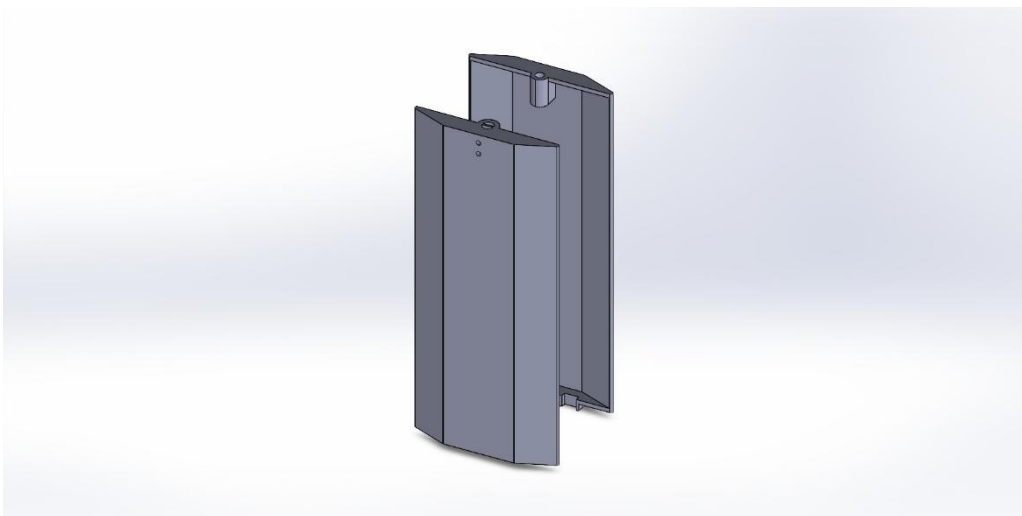


Figure 38 Trapezoidal Foils – “Trap” (Isometric View)

APPENDIX B: Pivot Tables and Graphs

This section highlights the results of the experiment using pivot tables and bar graphs to illustrate the relationship between the foil geometry and thickness with the performance metrics measured. This section will do a combined comparison for each performance metric, showing the data of all foils in both categories and how they compare in Strouhal number, power and efficiency. In the pivot tables, the maximum value of the performance metric for each variable, shape or thickness, is noted down as well.

Firstly, the pivot table and bar chart of the Strouhal number comparison can be observed. Here, the maximum Strouhal number observed across all factors is $St = 2.53$. The maximum Strouhal number of each shape can be seen in the last column while the maximum Strouhal number for each thickness, independent of geometry, can be seen in the last row.

Table 8 Strouhal Number Comparison Among All Shapes & Thickness indicating Max Strouhal Number Obtained (St)

Shapes	Strouhal Number					Max St across each Shape
	0	5	10	15	20	
flat	1.418492567	-	-	-	-	1.418492567
rect	1.418492567	1.789979826	1.619236228	1.452877863	1.341136707	1.789979826
roun	1.418492567	1.920900381	1.976517713	1.846484502	1.882799321	1.976517713
tria	1.418492567	2.087516567	2.085014576	2.021796739	1.896674577	2.087516567
conc	1.418492567	2.111783047	2.032157457	1.647178843	1.649439653	2.111783047
elli	1.418492567	2.530266794	2.271765008	1.985308558	1.855393241	2.530266794
circ	1.418492567	2.220874049	2.489964139	2.257894721	2.143157322	2.489964139
tri	1.418492567	2.323103086	2.371230255	2.332469096	1.962622995	2.371230255
trap	1.418492567	2.163902949	1.984348511	1.918626955	1.852494716	2.163902949
Max St across each thickness	1.418492567	2.530266794	2.489964139	2.332469096	2.143157322	2.530266794

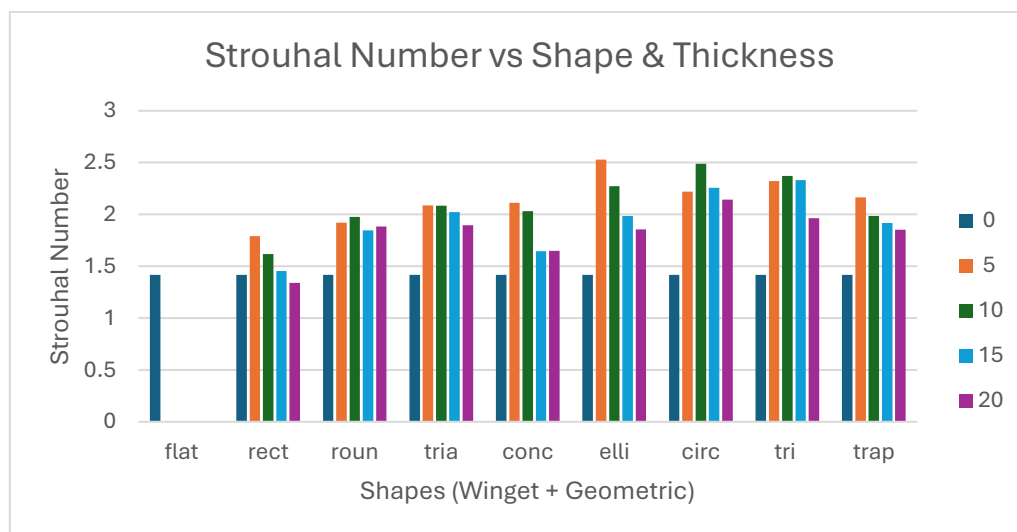


Figure 39 Bar Chart of Strouhal Number Comparison

Next, the pivot table and bar chart comparison of power across all foil shapes and thickness. Here, the maximum power across all factors is 0.113 W.

Table 9 Power Comparison among All Shapes & Thickness indicating Maximum Power Obtained

Shapes	Power					Max Power across each Shape
	0	5	10	15	20	
flat	0.069552 532	-	-	-	-	0.069552532
rect	0.069552 532	0.056901 723	0.0444906 82	0.038391 969	0.030923 886	0.069552532
roun	0.069552 532	0.087905 675	0.0839437 55	0.068137 928	0.073349 272	0.087905675
tria	0.069552 532	0.078381 156	0.0824945 44	0.072535 214	0.065222 397	0.082494544
conc	0.069552 532	0.087480 73	0.0756676 49	0.043996 85	0.045445 298	0.08748073
elli	0.069552 532	0.095047 654	0.0870225 45	0.075710 011	0.058964 122	0.095047654
circ	0.069552 532	0.081885 669	0.1126500 66	0.095140 829	0.086050 771	0.112650066
tri	0.069552 532	0.080779 451	0.0950229 22	0.089226 734	0.060859 297	0.095022922
trap	0.069552 532	0.070347 242	0.0659680 24	0.059862 86	0.055635 005	0.070347242
Max Power across each thickness	0.069552 532	0.095047 654	0.1126500 66	0.095140 829	0.086050 771	0.112650066

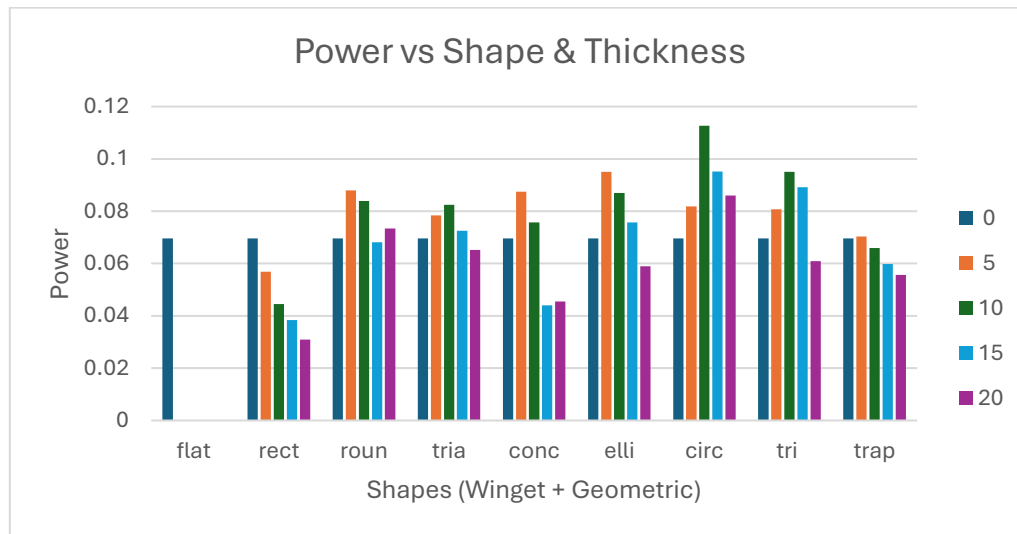


Figure 40 Bar Chart of Power (P) Comparison

Lastly, is the pivot table and bar chart for the efficiency comparison. Here, the maximum efficiency across all factors is $\eta = 0.453$ or 45%.

Table 10 Power Comparison among All Shapes & Thickness indicating Maximum Power Obtained

Shapes	Efficiency			Thickness (mm)		Max Efficiency across each Shape
	0	5	10	15	20	
flat	0.301740343	-	-	-	-	0.301740343
rect	0.301740343	0.246832016	0.187138305	0.159182922	0.126922493	0.301740343
roun	0.301740343	0.365635304	0.339825832	0.266785074	0.28007266	0.365635304
tria	0.301740343	0.335070891	0.342354812	0.292742662	0.257651699	0.342354812
conc	0.301740343	0.368893123	0.312619773	0.18181924	0.184802495	0.368893123
elli	0.301740343	0.399679876	0.351906985	0.298903561	0.229682398	0.399679876
circ	0.301740343	0.351346466	0.452707399	0.379381494	0.330183234	0.452707399
tri	0.301740343	0.343389102	0.387736146	0.358370157	0.246090337	0.387736146
trap	0.301740343	0.296974558	0.269901315	0.241317828	0.218472468	0.301740343
Max Efficiency across each thickness	0.301740343	0.399679876	0.452707399	0.379381494	0.330183234	0.452707399

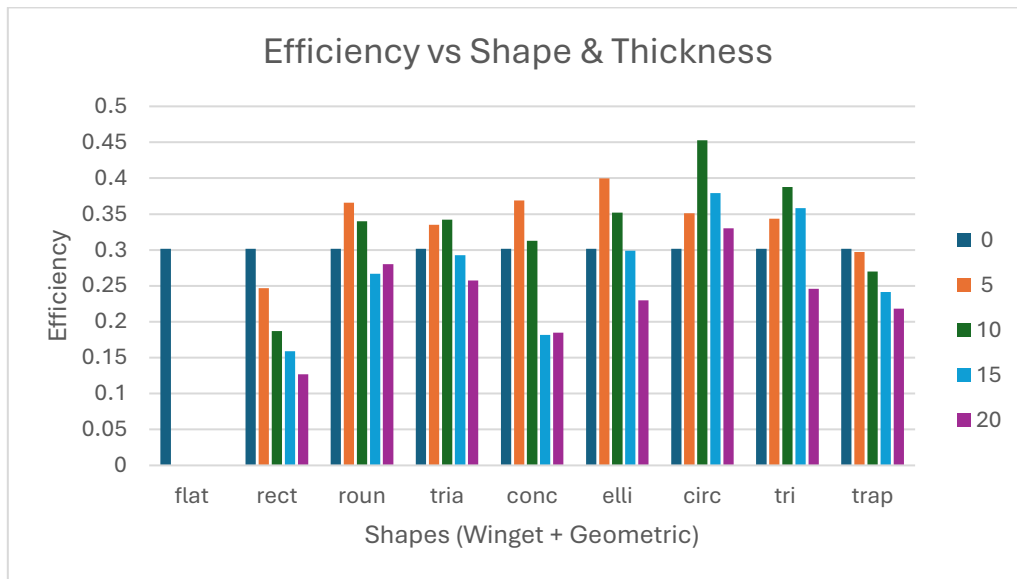


Figure 41 Bar Chart of Efficiency (η) Comparison